

Solaris Analytics

ADS508 | Group 7 | Authors: Luigi Salemi, Francisco Hernandez

This notebook covers ingestion and EDA for all primary datasets:

#	Dataset	Source	Type
1	Houston Solar Irradiance 2000–2019 (NREL NSRDB)	S3	CSV — irradiance + weather
2	GPVS-Faults PV Sensor Readings	S3	CSV — electrical sensor + fault labels
3	Solar Panel Images (<code>pythonafroz</code>)	KaggleHub	Images — clean vs faulty
4	PV Panel Defect Dataset (<code>alicjalena</code>)	KaggleHub	Images — 6 defect classes

S3 Bucket: `s3://solaris-analytics-bucket/`

GitHub: <https://github.com/darth-franky/Applied-Data-Science/tree/main/ADS508>

```
In [1]: import subprocess
subprocess.run(['pip', 'install', 'kagglehub', 'imbalanced-learn', '-q'])

import os
import json
import boto3
import sagemaker
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import seaborn as sns
import kagglehub
from pathlib import Path

from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report, confusion_matrix, f1_score
from imblearn.over_sampling import SMOTE

from sagemaker import get_execution_role, image_uris
from sagemaker.serializers import CSVSerializer
from sagemaker.inputs import TrainingInput
from sagemaker.estimator import Estimator

session = sagemaker.Session()
s3 = boto3.client('s3')
```

```

role = sagemaker.get_execution_role()

REGION = session.boto_region_name
BUCKET = 'solaris-analytics-bucket'
IMAGE_EXTENSIONS = {'.jpg', '.jpeg', '.png', '.bmp', '.tiff'}

print(f'SageMaker Role: {role}')
print(f'S3 Bucket: s3://{BUCKET}/')

```

sagemaker.config INFO – Not applying SDK defaults from location: /etc/xdg/sagemaker/config.yaml
 sagemaker.config INFO – Not applying SDK defaults from location: /home/sagemaker-user/.config/sagemaker/config.yaml
 SageMaker Role: arn:aws:iam::187806919439:role/LabRole
 S3 Bucket: s3://solaris-analytics-bucket/

```

In [2]: response = s3.list_objects_v2(Bucket=BUCKET)
objects = response.get('Contents', [])
print(f'Total objects in bucket: {len(objects)}')

```

Total objects in bucket: 1000

1. Ingest Houston Solar Irradiance Dataset (NREL NSRDB) from S3

File: Houston_Solar_2000_2019.csv

S3 path: s3://solaris-analytics-bucket/Houston_Solar_2000_2019.csv

Key fields: Timestamp, GHI, DNI, DHI, Temperature, Wind_Speed

Records: ~350,000 rows at 30-min intervals (2000-2019)

```

In [3]: import os

# Create folder
os.makedirs('/home/sagemaker-user/dataset1', exist_ok=True)

# Set paths
nsrdb_s3_key = 'Houston_Solar_2000_2019.csv'
nsrdb_local = '/home/sagemaker-user/dataset1/Houston_Solar_2000_2019.csv'

# Download from S3
print(f'Downloading s3://{BUCKET}/{nsrdb_s3_key} ...')
s3.download_file(BUCKET, nsrdb_s3_key, nsrdb_local)
print('Download complete.')

# Load into DataFrame
nsrdb_df = pd.read_csv(nsrdb_local, parse_dates=['Timestamp'])
print(f'Shape: {nsrdb_df.shape}')
nsrdb_df.head()

```

Downloading s3://solaris-analytics-bucket/Houston_Solar_2000_2019.csv ...
 Download complete.
 Shape: (350640, 6)

Out [3]:

	Timestamp	GHI	DNI	DHI	Temperature	Wind_Speed
0	2000-01-01 00:00:00	0	0	0	11.9	0.7
1	2000-01-01 00:30:00	0	0	0	11.5	0.7
2	2000-01-01 01:00:00	0	0	0	11.1	0.8
3	2000-01-01 01:30:00	0	0	0	10.6	0.9
4	2000-01-01 02:00:00	0	0	0	10.1	1.0

1.1 NSRDB — Basic Info & Missing Values

```
In [4]: print('=== Data Types ===')
print(nsrdb_df.dtypes)

print('\n=== Missing Values ===')
missing = nsrdb_df.isnull().sum()
print(missing[missing > 0] if missing.sum() > 0 else 'No missing values found')

print('\n=== Descriptive Statistics ===')
display(nsrdb_df.describe().T)
```

=== Data Types ===

Timestamp	datetime64[ns]
GHI	int64
DNI	int64
DHI	int64
Temperature	float64
Wind_Speed	float64
dtype:	object

=== Missing Values ===

No missing values found.

=== Descriptive Statistics ===

	count	mean	min	25%	50%	75%	
Timestamp	350640	2009-12-31 11:45:00.000001024	2000-01-01 00:00:00	2004-12-31 05:52:30	2009-12-31 11:45:00	2014-12-31 17:37:30	2014-12-31 23:30:00
GHI	350640.0	201.67449	0.0	0.0	0.0	365.0	1000.0
DNI	350640.0	197.567154	0.0	0.0	0.0	348.0	1000.0
DHI	350640.0	79.42834	0.0	0.0	0.0	120.0	600.0
Temperature	350640.0	20.82322	-7.0	15.2	22.2	26.6	
Wind_Speed	350640.0	2.218328	0.2	1.3	2.0	2.9	

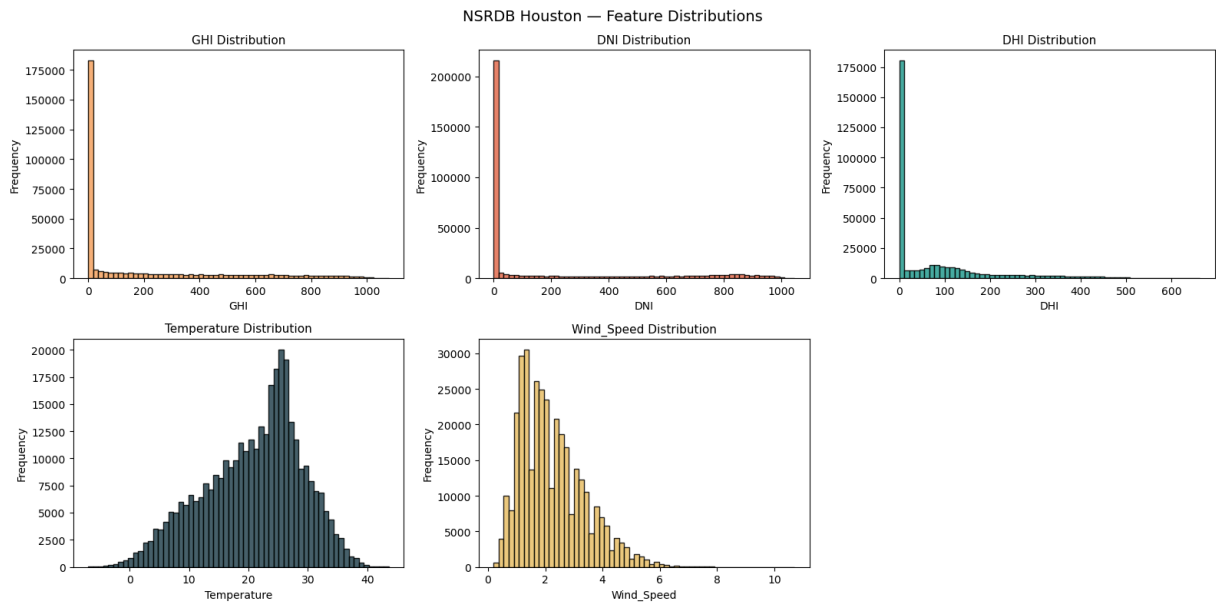
1.2 NSRDB — Irradiance & Weather Distributions

```
In [5]: irradiance_cols = ['GHI', 'DNI', 'DHI']
weather_cols = ['Temperature', 'Wind_Speed']
all_numeric = irradiance_cols + weather_cols

fig, axes = plt.subplots(2, 3, figsize=(16, 8))
axes = axes.flatten()
colors = ['#F4A261', '#E76F51', '#2A9D8F', '#264653', '#E9C46A']

for i, col in enumerate(all_numeric):
    axes[i].hist(nsrdb_df[col].dropna(), bins=60, color=colors[i], edgecolor='black')
    axes[i].set_title(f'{col} Distribution', fontsize=11)
    axes[i].set_xlabel(col)
    axes[i].set_ylabel('Frequency')

axes[5].set_visible(False)
plt.suptitle('NSRDB Houston - Feature Distributions', fontsize=14)
plt.tight_layout()
plt.savefig('/tmp/nsrdb_distributions.png', dpi=150)
plt.show()
print('Note: High frequency of GHI=0 reflects nighttime - these rows may be
```



Note: High frequency of GHI=0 reflects nighttime - these rows may be filtered before forecasting.

1.3 NSRDB — Seasonal & Temporal Bias Check

```
In [6]: nsrdb_df['Year'] = nsrdb_df['Timestamp'].dt.year
nsrdb_df['Month'] = nsrdb_df['Timestamp'].dt.month
nsrdb_df['Hour'] = nsrdb_df['Timestamp'].dt.hour

fig, axes = plt.subplots(1, 3, figsize=(18, 4))

nsrdb_df['Year'].value_counts().sort_index().plot(
    kind='bar', ax=axes[0], color='steelblue', edgecolor='black')
axes[0].set_title('Records per Year')
axes[0].set_xlabel('Year')
axes[0].set_ylabel('Count')
```

```

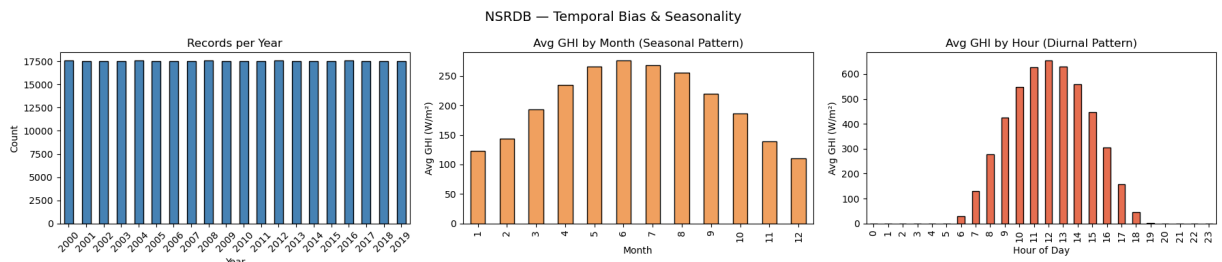
axes[0].tick_params(axis='x', rotation=45)

nsrdb_df.groupby('Month')['GHI'].mean().plot(
    kind='bar', ax=axes[1], color='#F4A261', edgecolor='black')
axes[1].set_title('Avg GHI by Month (Seasonal Pattern)')
axes[1].set_xlabel('Month')
axes[1].set_ylabel('Avg GHI (W/m²)')

nsrdb_df.groupby('Hour')['GHI'].mean().plot(
    kind='bar', ax=axes[2], color='#E76F51', edgecolor='black')
axes[2].set_title('Avg GHI by Hour (Diurnal Pattern)')
axes[2].set_xlabel('Hour of Day')
axes[2].set_ylabel('Avg GHI (W/m²)')

plt.suptitle('NSRDB – Temporal Bias & Seasonality', fontsize=14)
plt.tight_layout()
plt.savefig('/tmp/nsrdb_temporal_bias.png', dpi=150)
plt.show()
print('Note: NSRDB is U.S.-centric (Houston only). Seasonal and diurnal patt

```



Note: NSRDB is U.S.-centric (Houston only). Seasonal and diurnal patterns are expected physical behavior.

1.4 NSRDB — GHI Irradiance Category Distribution

Per the design doc, GHI will be bucketed into **Low / Medium / High** for classification tasks.

```

In [7]: daytime_df = nsrdb_df[nsrdb_df['GHI'] > 0].copy()
daytime_df['GHI_Category'] = pd.cut(
    daytime_df['GHI'],
    bins=[0, 200, 600, daytime_df['GHI'].max() + 1],
    labels=['Low', 'Medium', 'High']
)

print('=== GHI Category Distribution (daytime only) ===')
print(daytime_df['GHI_Category'].value_counts())

fig, ax = plt.subplots(figsize=(6, 4))
daytime_df['GHI_Category'].value_counts().reindex(['Low', 'Medium', 'High']).plot(
    kind='bar', ax=ax, color=['#2A9D8F', '#E9C46A', '#E76F51'], edgecolor='black')
ax.set_title('GHI Irradiance Category Distribution (Daytime Only)', fontsize=14)
ax.set_xlabel('Category')
ax.set_ylabel('Count')
ax.tick_params(axis='x', rotation=0)
plt.tight_layout()

```

```
plt.savefig('/tmp/nsrdb_ghi_categories.png', dpi=150)
plt.show()
```

=== GHI Category Distribution (daytime only) ===

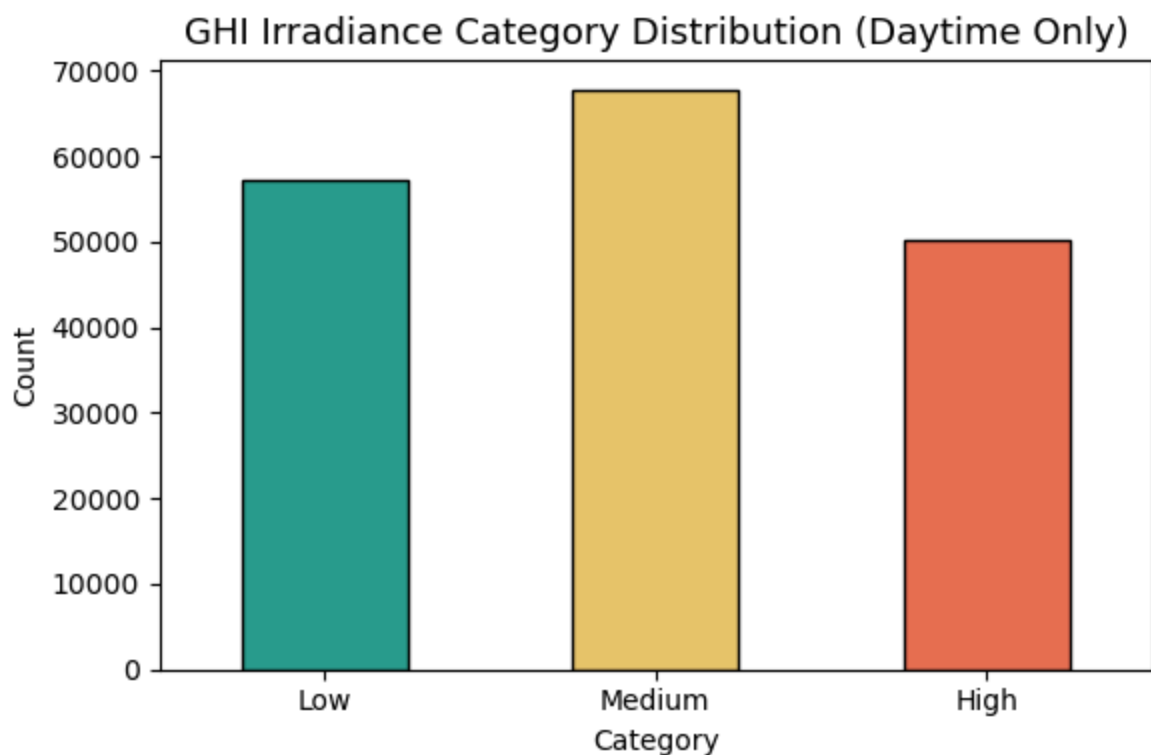
GHI_Category

Medium 67793

Low 57143

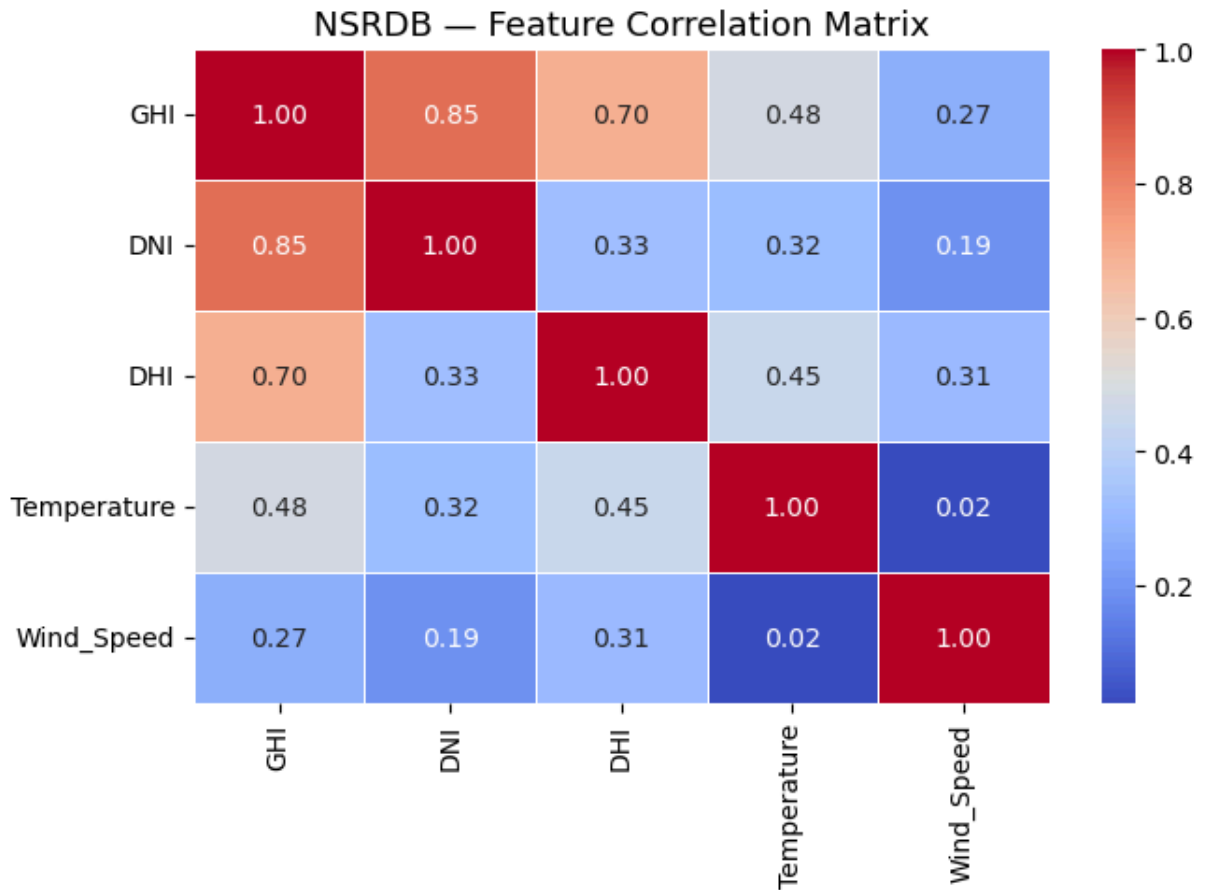
High 50195

Name: count, dtype: int64



1.5 NSRDB — Correlation Heatmap

```
In [8]: fig, ax = plt.subplots(figsize=(7, 5))
sns.heatmap(nsrdb_df[all_numeric].corr(), annot=True, fmt='.2f', cmap='coolw
ax.set_title('NSRDB - Feature Correlation Matrix', fontsize=13)
plt.tight_layout()
plt.savefig('/tmp/nsrdb_correlation.png', dpi=150)
plt.show()
```



2. Ingest GPVS-Faults Sensor Dataset from S3

S3 path: `s3://solaris-analytics-bucket/dataset2/gpvs-faults/`

Key fields: voltage, current, power, fault_label

Records: 10,000+

```
In [9]: response = s3.list_objects_v2(Bucket=BUCKET)
gpvs_keys = [obj['Key'] for obj in response.get('Contents', [])]

local_gpvs_dir = '/home/sagemaker-user/dataset2/gpvs-faults/'
os.makedirs(local_gpvs_dir, exist_ok=True)

for key in gpvs_keys:
    filename = key.split('/')[-1]
    if filename:
        local_path = os.path.join(local_gpvs_dir, filename)
        s3.download_file(BUCKET, key, local_path)
        #print(f'Downloaded: {key}')

csv_files = list(Path(local_gpvs_dir).glob('*.csv')) + list(Path(local_gpvs_dir).glob('*.tsv'))
dfs = []
for f in csv_files:
    sep = '\t' if f.suffix == '.tsv' else ','
    df_temp = pd.read_csv(f, sep=sep)
    df_temp['source_file'] = f.name
    dfs.append(df_temp)
```

```
print(f'Loaded {f.name}: {df_temp.shape}')
```

```
gpvs_df = pd.concat(dfs, ignore_index=True)
print(f'\nCombined GPVS shape: {gpvs_df.shape}')
```

```
gpvs_df.head()
```

```
Loaded F0L.csv: (143715, 15)
Loaded F0M.csv: (141014, 15)
Loaded F1L.csv: (129013, 15)
Loaded F1M.csv: (139014, 15)
Loaded F2L.csv: (142128, 15)
Loaded F2M.csv: (144015, 15)
Loaded F3L.csv: (103497, 15)
Loaded F3M.csv: (69967, 15)
Loaded F4L.csv: (144014, 15)
Loaded F4M.csv: (144014, 15)
Loaded F5L.csv: (143015, 15)
Loaded F5M.csv: (144014, 15)
Loaded F6L.csv: (144015, 15)
Loaded F6M.csv: (144015, 15)
Loaded F7L.csv: (144015, 15)
Loaded F7M.csv: (144015, 15)
Loaded Houston_Solar_2000_2019.csv: (350640, 7)
```

```
Combined GPVS shape: (2514120, 21)
```

Out[9]:

	Time	Ipv	Vpv	Vdc	ia	ib	ic	
0	0.000028	1.572327	101.348877	144.140625	-0.135133	0.490112	-0.354985	41.7
1	0.000128	1.503265	101.458740	143.554688	-0.108277	0.510254	-0.388555	46.8
2	0.000228	1.492859	101.574707	143.554688	-0.168702	0.496826	-0.334844	51.0
3	0.000328	1.558136	101.312256	143.261719	-0.135133	0.510254	-0.361699	55.8
4	0.000428	1.631927	101.141357	143.847656	-0.202271	0.503540	-0.321416	60.0

5 rows x 21 columns

2.1 GPVS — Basic Info, Missing Values & Zero-Value Check

```
In [10]: print('=== Data Types ===')
print(gpvs_df.dtypes)

print('\n=== Missing Values ===')
missing = gpvs_df.isnull().sum()
print(missing[missing > 0] if missing.sum() > 0 else 'No missing values found')

numeric_cols = gpvs_df.select_dtypes(include=[np.number]).columns.tolist()

print('\n=== Zero-Value Counts (potential sensor outage indicator) ===')
zero_counts = (gpvs_df[numeric_cols] == 0).sum()
zero_pct = (zero_counts / len(gpvs_df) * 100).round(2)
```



```
print(pd.DataFrame({'Zero Count': zero_counts, 'Zero %': zero_pct}).to_string())  
print('\nNote: High zero % in voltage/current/power may indicate sensor outa
```

=== Data Types ===

Time	float64
Ipv	float64
Vpv	float64
Vdc	float64
ia	float64
ib	float64
ic	float64
va	float64
vb	float64
vc	float64
Iabc	float64
If	float64
Vabc	float64
Vf	float64
source_file	object
Timestamp	object
GHI	float64
DNI	float64
DHI	float64
Temperature	float64
Wind_Speed	float64
dtype:	object

=== Missing Values ===

Time	350640
Ipv	350640
Vpv	350640
Vdc	350640
ia	350640
ib	350640
ic	350640
va	350640
vb	350640
vc	350640
Iabc	350640
If	350640
Vabc	350640
Vf	350640
Timestamp	2163480
GHI	2163480
DNI	2163480
DHI	2163480
Temperature	2163480
Wind_Speed	2163480
dtype:	int64

=== Zero-Value Counts (potential sensor outage indicator) ===

	Zero Count	Zero %
Time	0	0.00
Ipv	48	0.00
Vpv	0	0.00
Vdc	1	0.00
ia	0	0.00
ib	21441	0.85
ic	0	0.00

va	55	0.00
vb	55	0.00
vc	0	0.00
Iabc	0	0.00
If	0	0.00
Vabc	0	0.00
Vf	0	0.00
GHI	175509	6.98
DNI	195739	7.79
DHI	175509	6.98
Temperature	90	0.00
Wind_Speed	0	0.00

Note: High zero % in voltage/current/power may indicate sensor outages needing removal in data prep.

2.2 GPVS — Feature Distributions & Correlation

```
In [11]: plot_cols = [c for c in numeric_cols if c in
                    ['Ipv', 'Vpv', 'Vdc', 'ia', 'ib', 'ic', 'va', 'vb', 'vc', 'Iabc', 'If', 'V

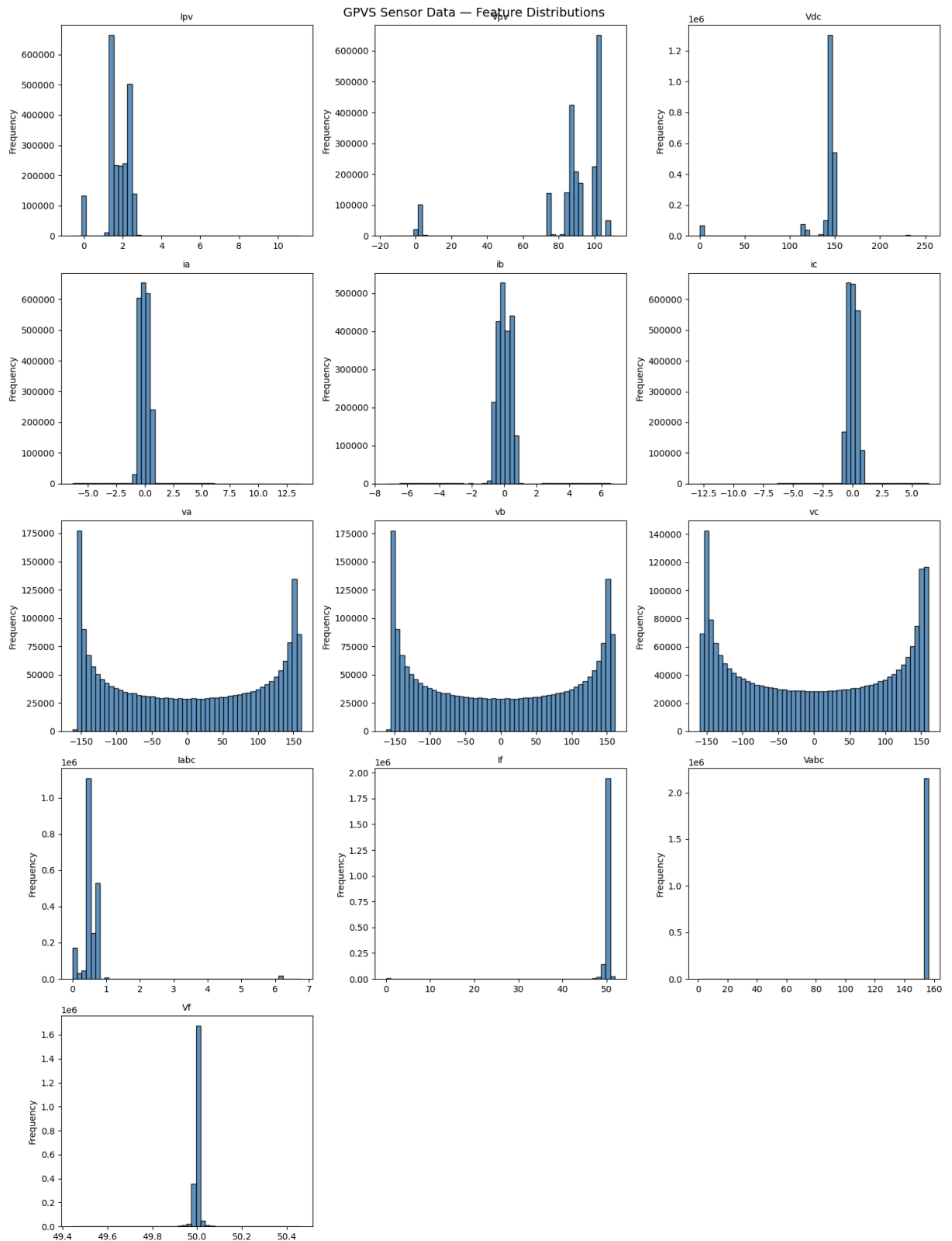
n = len(plot_cols)
cols_per_row = 3
rows = (n + cols_per_row - 1) // cols_per_row

fig, axes = plt.subplots(rows, cols_per_row, figsize=(15, rows * 4))
axes = axes.flatten()

for i, col in enumerate(plot_cols):
    axes[i].hist(gpvs_df[col].dropna(), bins=50, color='steelblue', edgecolor='black')
    axes[i].set_title(col, fontsize=10)
    axes[i].set_ylabel('Frequency')

for j in range(i + 1, len(axes)):
    axes[j].set_visible(False)

plt.suptitle('GPVS Sensor Data — Feature Distributions', fontsize=14)
plt.tight_layout()
plt.savefig('/tmp/gpvs_distributions.png', dpi=150)
plt.show()
```



3. Ingest Image Datasets via KaggleHub → Upload to S3

Two image datasets downloaded from Kaggle and persisted to S3:

- **pythonafroz/solar-panel-images** → binary: clean vs faulty panels
- **alicjalena/pv-panel-defect-dataset** → multi-class: 6 defect types

```
In [12]: def upload_dir_to_s3(local_dir, s3_prefix, bucket):
    """Recursively upload a local directory to an S3 prefix."""
    count = 0
    for root, _, files in os.walk(local_dir):
        for file in files:
            local_path = os.path.join(root, file)
            rel_path = os.path.relpath(local_path, local_dir)
            s3_key = s3_prefix + rel_path.replace(os.sep, '/')
            s3.upload_file(local_path, bucket, s3_key)
            count += 1
    return count

def count_images_by_class(base_path):
    """Return dict of {class_name: image_count} from a folder of class subfolders"""
    counts = {}
    for item in Path(base_path).iterdir():
        if item.is_dir():
            n = sum(1 for f in item.rglob('*') if f.suffix.lower() in IMAGE_SUFFIXES)
            if n > 0:
                counts[item.name] = n
    return counts

print('Helper functions defined.')
```

Helper functions defined.

```
In [13]: # Download Solar Panel Images (pythonafroz)
print('Downloading pythonafroz/solar-panel-images ...')
solar_panel_path = kagglehub.dataset_download('pythonafroz/solar-panel-images')
print(f'Downloaded to: {solar_panel_path}')

n = upload_dir_to_s3(solar_panel_path, 'panel-images/solar-panel-images/', BUCKET)
print(f'Uploaded {n} files to s3://{BUCKET}/panel-images/solar-panel-images/')
```

Downloading pythonafroz/solar-panel-images ...

Downloaded to: /home/sagemaker-user/.cache/kagglehub/datasets/pythonafroz/solar-panel-images/versions/2

Uploaded 891 files to s3://solaris-analytics-bucket/panel-images/solar-panel-images/

```
In [14]: # Download PV Panel Defect Dataset (alicjalena)
print('Downloading alicjalena/pv-panel-defect-dataset ...')
pv_defect_path = kagglehub.dataset_download('alicjalena/pv-panel-defect-dataset')
print(f'Downloaded to: {pv_defect_path}')

n = upload_dir_to_s3(pv_defect_path, 'panel-images/pv-defect/', BUCKET)
print(f'Uploaded {n} files to s3://{BUCKET}/panel-images/pv-defect/')
```

Downloading alicjalena/pv-panel-defect-dataset ...

Downloaded to: /home/sagemaker-user/.cache/kagglehub/datasets/alicjalena/pv-panel-defect-dataset/versions/1

Uploaded 1574 files to s3://solaris-analytics-bucket/panel-images/pv-defect/

3.1 Solar Panel Images (pythonafroz) — Distribution & Samples

```
In [15]: solar_counts = count_images_by_class(solar_panel_path)
print('=== Solar Panel Images – Images per Class ===')
for cls, cnt in sorted(solar_counts.items()):
    print(f' {cls}: {cnt} images')
print(f'Total: {sum(solar_counts.values())}')

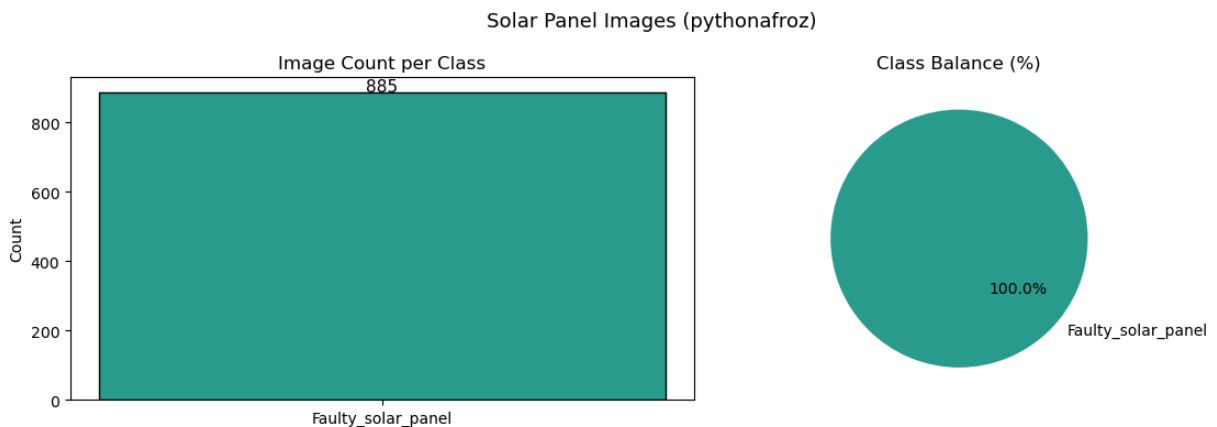
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

axes[0].bar(solar_counts.keys(), solar_counts.values(), color=['#2A9D8F', '#E76F51'])
for i, (cls, cnt) in enumerate(solar_counts.items()):
    axes[0].text(i, cnt + 5, str(cnt), ha='center', fontsize=11)
axes[0].set_title('Image Count per Class', fontsize=12)
axes[0].set_ylabel('Count')

axes[1].pie(solar_counts.values(), labels=solar_counts.keys(),
            autopct='%1.1f%%', colors=['#2A9D8F', '#E76F51'], startangle=140)
axes[1].set_title('Class Balance (%)', fontsize=12)

plt.suptitle('Solar Panel Images (pythonafroz)', fontsize=13)
plt.tight_layout()
plt.savefig('/tmp/solar_panel_distribution.png', dpi=150)
plt.show()
```

```
=== Solar Panel Images – Images per Class ===
Faulty_solar_panel: 885 images
Total: 885
```



```
In [16]: # Sample one image per class
classes = sorted(solar_counts.keys())
fig, axes = plt.subplots(1, len(classes), figsize=(6 * len(classes), 5))
if len(classes) == 1:
    axes = [axes]

for ax, cls in zip(axes, classes):
    class_dir = Path(solar_panel_path) / cls
    imgs = [f for f in class_dir.rglob('*') if f.suffix.lower() in IMAGE_EXT]
    if imgs:
        ax.imshow(mpimg.imread(str(imgs[0])))
        ax.set_title(cls, fontsize=12)
        ax.axis('off')
```

```
plt.suptitle('Solar Panel Images – Sample per Class', fontsize=13)
plt.tight_layout()
plt.savefig('/tmp/solar_panel_samples.png', dpi=150)
plt.show()
```

Solar Panel Images — Sample per Class

Faulty_solar_panel



3.2 PV Panel Defect Dataset (alicjalena) — Distribution & Samples

```
In [17]: defect_counts = count_images_by_class(pv_defect_path)
print('=== PV Panel Defect Dataset – Images per Class ===')
for cls, cnt in sorted(defect_counts.items()):
    print(f' {cls}: {cnt} images')
print(f'Total: {sum(defect_counts.values())}')

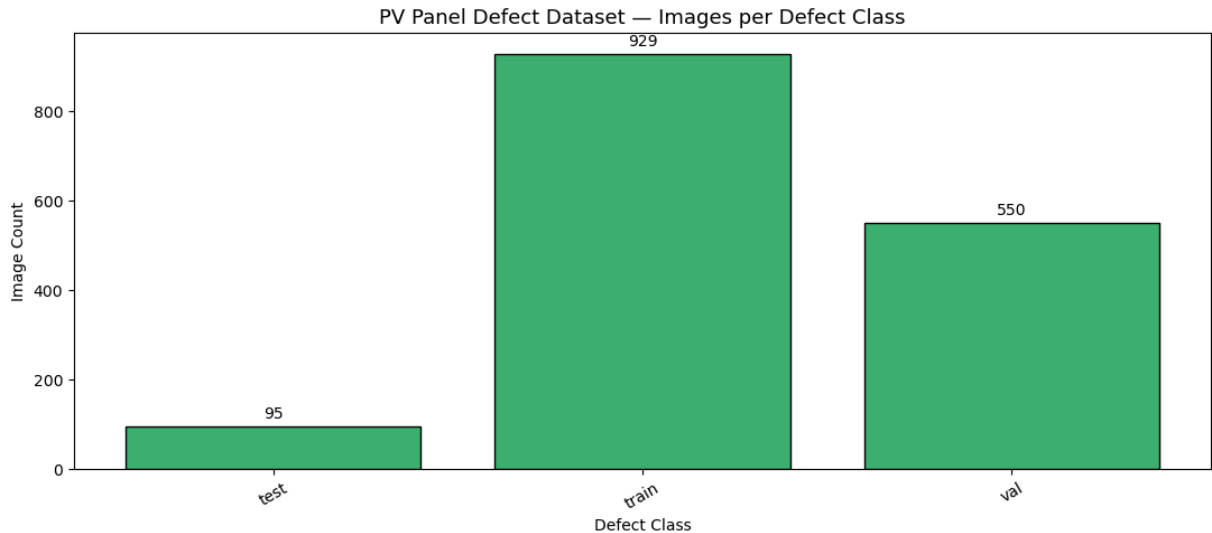
fig, ax = plt.subplots(figsize=(11, 5))
bars = ax.bar(defect_counts.keys(), defect_counts.values(), color='mediumseagreen')
ax.bar_label(bars, padding=3)
ax.set_title('PV Panel Defect Dataset – Images per Defect Class', fontsize=13)
ax.set_xlabel('Defect Class')
ax.set_ylabel('Image Count')
ax.tick_params(axis='x', rotation=30)
plt.tight_layout()
plt.savefig('/tmp/pv_defect_distribution.png', dpi=150)
```

```
plt.show()
print('Note: Underrepresented defect classes will require augmentation or cl
```

=== PV Panel Defect Dataset – Images per Class ===

test: 95 images
train: 929 images
val: 550 images

Total: 1574



Note: Underrepresented defect classes will require augmentation or class weighting during CNN training.

```
In [18]: display_classes = sorted(defect_counts.keys())[:6]
fig, axes = plt.subplots(1, len(display_classes), figsize=(3 * len(display_classes), 3))
if len(display_classes) == 1:
    axes = [axes]

for ax, cls in zip(axes, display_classes):
    class_dir = Path(pv_defect_path) / cls
    imgs = [f for f in class_dir.rglob('*') if f.suffix.lower() in IMAGE_EXTENSIONS]
    if imgs:
        ax.imshow(mplimg.imread(str(imgs[0])))
        ax.set_title(cls, fontsize=9)
        ax.axis('off')

plt.suptitle('PV Panel Defect Dataset – Sample per Defect Class', fontsize=12)
plt.tight_layout()
plt.savefig('/tmp/pv_defect_samples.png', dpi=150)
plt.show()
```


PV Panel Defect Dataset — Sample per Defect Class



4. Full EDA Summary

```
In [19]: print('=' * 62)
print('      SOLARIS ANALYTICS – COMPLETE EDA SUMMARY')
print('=' * 62)

print(f'\n[1] NSRDB Houston Solar Irradiance (2000–2019)')
print(f'    Rows:      {len(nsrdb_df):,}')
print(f'    Columns:    {nsrdb_df.shape[1]}')
print(f'    Date Range: {nsrdb_df["Timestamp"].min()} to {nsrdb_df["Timestamp"].max()}')
print(f'    Missing:    {nsrdb_df.isnull().sum().sum()}')
print(f'    GHI Range:  {nsrdb_df["GHI"].min()} – {nsrdb_df["GHI"].max()} W/m²')
print(f'    Key Risk:   U.S.–centric (Houston only); seasonal/diurnal bias e

print(f'\n[2] GPVS–Faults Sensor Data')
print(f'    Rows:      {len(gpvs_df):,}')
print(f'    Columns:    {gpvs_df.shape[1]}')
print(f'    Missing:    {gpvs_df.isnull().sum().sum()}')
print(f'    Key Risk:   Single PV system source; check fault class imbalance

print(f'\n[3] Solar Panel Images (pythonafroz) – Binary Classification')
for cls, cnt in sorted(solar_counts.items()):
    print(f'    {cls}: {cnt} images')
print(f'    Total: {sum(solar_counts.values())} | Key Risk: Binary class bal

print(f'\n[4] PV Panel Defect Dataset (alicjalena) – Multi-class (6 classes)')
for cls, cnt in sorted(defect_counts.items()):
    print(f'    {cls}: {cnt} images')
print(f'    Total: {sum(defect_counts.values())} | Key Risk: Multi-class imb

print(f'\nAll data stored at: s3://{BUCKET}/')
print(f'    nsrdb/              <- NREL irradiance CSV')
print(f'    gpvs-faults/        <- PV sensor fault CSV')
print(f'    panel-images/solar-panel-images/ <- pythonafroz images')
print(f'    panel-images/pv-defect/         <- alicjalena images')
```

```
print(f'\nGitHub: https://github.com/darth-franky/Applied-Data-Science/tree/main/ADS508\n')
print('=' * 62)
```

SOLARIS ANALYTICS – COMPLETE EDA SUMMARY

- [1] NSRDB Houston Solar Irradiance (2000–2019)
 - Rows: 350,640
 - Columns: 9
 - Date Range: 2000–01–01 00:00:00 to 2019–12–31 23:30:00
 - Missing: 0
 - GHI Range: 0 – 1078 W/m2
 - Key Risk: U.S.-centric (Houston only); seasonal/diurnal bias expected
- [2] GPVS–Faults Sensor Data
 - Rows: 2,514,120
 - Columns: 21
 - Missing: 17889840
 - Key Risk: Single PV system source; check fault class imbalance
- [3] Solar Panel Images (pythonafroz) – Binary Classification
 - Faulty_solar_panel: 885 images
 - Total: 885 | Key Risk: Binary class balance
- [4] PV Panel Defect Dataset (alicjalena) – Multi-class (6 classes)
 - test: 95 images
 - train: 929 images
 - val: 550 images
 - Total: 1574 | Key Risk: Multi-class imbalance

All data stored at: s3://solaris-analytics-bucket/

- nsrdb/ <- NREL irradiance CSV
- gpvs-faults/ <- PV sensor fault CSV
- panel-images/solar-panel-images/ <- pythonafroz images
- panel-images/pv-defect/ <- alicjalena images

GitHub: <https://github.com/darth-franky/Applied-Data-Science/tree/main/ADS508>

2. Data Scrubbing — NSRDB Irradiance

Cleansing steps applied:

- No missing values found in EDA — confirmed below
- Remove physically impossible values (negative irradiance)
- Filter to daytime only (GHI > 0) for forecasting model
- Extract temporal features from Timestamp

```
In [20]: print(f'NSRDB rows (raw): {len(nsrdb_df),}')
print(f'Missing values: {nsrdb_df.isnull().sum().sum()}')
```

```
# Remove negative irradiance (physically impossible)
nsrdb_clean = nsrdb_df.copy()
for col in ['GHI', 'DNI', 'DHI']:
    nsrdb_clean = nsrdb_clean[nsrdb_clean[col] >= 0]

# Filter to daytime only for forecasting (nighttime GHI=0 adds noise)
nsrdb_daytime = nsrdb_clean[nsrdb_clean['GHI'] > 0].copy()

print(f'After cleaning:      {len(nsrdb_clean):,} rows')
print(f'Daytime only:        {len(nsrdb_daytime):,} rows')
print(f'Nighttime removed:{len(nsrdb_clean) - len(nsrdb_daytime):,} rows')
```

```
NSRDB rows (raw): 350,640
Missing values:    0
After cleaning:    350,640 rows
Daytime only:      175,131 rows
Nighttime removed:175,509 rows
```

2.1 NSRDB — Feature Engineering

Feature creation: Extract temporal features from Timestamp and apply cyclical encoding to preserve periodicity (month 12 → month 1, hour 23 → hour 0).

Feature selection: Use DNI, DHI, Temperature, Wind_Speed + engineered temporal features. Exclude raw Timestamp, Month, Hour (replaced by cyclical versions).

```
In [21]: nsrdb_daytime['Month']      = nsrdb_daytime['Timestamp'].dt.month
nsrdb_daytime['Hour']      = nsrdb_daytime['Timestamp'].dt.hour
nsrdb_daytime['DayOfYear'] = nsrdb_daytime['Timestamp'].dt.dayofyear

# Cyclical encoding – prevents model from treating Dec->Jan as a large jump
nsrdb_daytime['Month_sin'] = np.sin(2 * np.pi * nsrdb_daytime['Month'] / 12)
nsrdb_daytime['Month_cos'] = np.cos(2 * np.pi * nsrdb_daytime['Month'] / 12)
nsrdb_daytime['Hour_sin']  = np.sin(2 * np.pi * nsrdb_daytime['Hour'] / 24)
nsrdb_daytime['Hour_cos']  = np.cos(2 * np.pi * nsrdb_daytime['Hour'] / 24)

# GHI category buckets (Low/Medium/High) per design doc
nsrdb_daytime['GHI_Category'] = pd.cut(
    nsrdb_daytime['GHI'],
    bins=[0, 200, 600, nsrdb_daytime['GHI'].max() + 1],
    labels=[0, 1, 2] # 0=Low, 1=Medium, 2=High
).astype(int)

print('GHI Category distribution (daytime):')
print(nsrdb_daytime['GHI_Category'].value_counts().sort_index())
print('\n0=Low (<200), 1=Medium (200-600), 2=High (>600)')
```

GHI Category distribution (daytime):

GHI_Category

0 57143

1 67793

2 50195

Name: count, dtype: int64

0=Low (<200), 1=Medium (200–600), 2=High (>600)

2.2 NSRDB — Feature Selection & Scaling

```
In [22]: # Features: physical weather + cyclical temporal
NSRDB_FEATURES = ['DNI', 'DHI', 'Temperature', 'Wind_Speed',
                  'Month_sin', 'Month_cos', 'Hour_sin', 'Hour_cos', 'DayOfYe
NSRDB_TARGET    = 'GHI'

X_nsrdb = nsrdb_daytime[NSRDB_FEATURES]
y_nsrdb = nsrdb_daytime[NSRDB_TARGET]

scaler_nsrdb = StandardScaler()
X_nsrdb_scaled = pd.DataFrame(
    scaler_nsrdb.fit_transform(X_nsrdb),
    columns=NSRDB_FEATURES
)

print(f'Features: {NSRDB_FEATURES}')
print(f'Target:    {NSRDB_TARGET}')
print(f'Shape:      {X_nsrdb_scaled.shape}')
```

Features: ['DNI', 'DHI', 'Temperature', 'Wind_Speed', 'Month_sin', 'Month_co
s', 'Hour_sin', 'Hour_cos', 'DayOfYear']

Target: GHI

Shape: (175131, 9)

2.3 NSRDB — Train / Validation / Test Split (70/15/15)

```
In [23]: X_train_n, X_temp_n, y_train_n, y_temp_n = train_test_split(
    X_nsrdb_scaled, y_nsrdb, test_size=0.30, random_state=42)
X_val_n, X_test_n, y_val_n, y_test_n = train_test_split(
    X_temp_n, y_temp_n, test_size=0.50, random_state=42)

total = len(X_nsrdb_scaled)
print('NSRDB Split:')
print(f' Train:      {len(X_train_n):,} ({len(X_train_n)/total*100:.1f}%)')
print(f' Validation: {len(X_val_n):,} ({len(X_val_n)/total*100:.1f}%)')
print(f' Test:       {len(X_test_n):,} ({len(X_test_n)/total*100:.1f}%)')
```

NSRDB Split:

Train: 122,591 (70.0%)

Validation: 26,270 (15.0%)

Test: 26,270 (15.0%)

3. Data Scrubbing — GPVS Fault Sensor Data

Key findings from EDA:

- 16 scenario files (F0L, F0M ... F7L, F7M) — fault type encoded in filename
- Actual sensor columns: `Ipv`, `Vpv`, `Vdc`, `ia`, `ib`, `ic`, `va`, `vb`, `vc`, `Iabc`, `If`, `Vabc`, `Vf`
- `Time` is a relative timestamp within each scenario, not a real datetime
- `GHI/DNI/DHI/Timestamp` columns present from accidental merge — **excluded**

Cleansing steps:

- Drop accidental NSRDB columns
- Drop rows with missing sensor values
- Use `fault_type` (F0–F7) as classification target — 8 classes

```
In [24]: print(f'GPVS rows (raw): {len(gpvs_df),}')
print(f'Columns: {gpvs_df.columns.tolist()}')

# Drop columns that came from the accidental NSRDB merge
accidental_cols = ['Timestamp', 'GHI', 'DNI', 'DHI', 'Temperature', 'Wind_Speed']
accidental_cols = [c for c in accidental_cols if c in gpvs_df.columns]
gpvs_clean = gpvs_df.drop(columns=accidental_cols).copy()
print(f'\nDropped accidental columns: {accidental_cols}')
print(f'Remaining columns: {gpvs_clean.columns.tolist()}')
```

GPVS rows (raw): 2,514,120

Columns: ['Time', 'Ipv', 'Vpv', 'Vdc', 'ia', 'ib', 'ic', 'va', 'vb', 'vc', 'Iabc', 'If', 'Vabc', 'Vf', 'source_file', 'Timestamp', 'GHI', 'DNI', 'DHI', 'Temperature', 'Wind_Speed']

Dropped accidental columns: ['Timestamp', 'GHI', 'DNI', 'DHI', 'Temperature', 'Wind_Speed', 'source_file']

Remaining columns: ['Time', 'Ipv', 'Vpv', 'Vdc', 'ia', 'ib', 'ic', 'va', 'vb', 'vc', 'Iabc', 'If', 'Vabc', 'Vf']

```
In [25]: local_gpvs_dir = '/home/sagemaker-user/dataset2/gpvs-faults/'
csv_files = list(Path(local_gpvs_dir).glob('*.csv'))

# Columns that don't belong in GPVS data
ACCIDENTAL_COLS = ['Timestamp', 'GHI', 'DNI', 'DHI', 'Temperature', 'Wind_Speed']

dfs = []
for f in csv_files:
    df_temp = pd.read_csv(f)
    df_temp = df_temp.drop(columns=[c for c in ACCIDENTAL_COLS if c in df_temp.columns])
    df_temp['scenario'] = f.stem
    df_temp['fault_type'] = f.stem[:1]
    df_temp['fault_severity'] = f.stem[-1]
    dfs.append(df_temp)

gpvs_df = pd.concat(dfs, ignore_index=True)
gpvs_clean = gpvs_df.dropna().copy()
```

```
print(f'GPVS loaded: {gpvs_df.shape}')
print(f'GPVS cleaned: {gpvs_clean.shape}')
print(f'Fault types: {sorted(gpvs_clean["fault_type"].unique())}')
```

GPVS loaded: (2514120, 17)

GPVS cleaned: (2163480, 17)

Fault types: ['F0', 'F1', 'F2', 'F3', 'F4', 'F5', 'F6', 'F7']

3.1 GPVS — Feature Selection

Selected features: Core electrical sensor readings: Ipv, Vpv, Vdc, ia, ib, ic, va, vb, vc, Iabc, If, Vabc, Vf

Excluded: Time (relative timestamp, not a real feature), scenario, fault_severity (keep only fault type as target)

Target: fault_type (F0–F7, 8 classes)

```
In [26]: GPVS_SENSOR_COLS = ['Ipv', 'Vpv', 'Vdc', 'ia', 'ib', 'ic',
                             'va', 'vb', 'vc', 'Iabc', 'If', 'Vabc', 'Vf']

# Verify all expected columns exist
missing_cols = [c for c in GPVS_SENSOR_COLS if c not in gpvs_clean.columns]
if missing_cols:
    print(f'WARNING: Missing columns: {missing_cols}')
    GPVS_SENSOR_COLS = [c for c in GPVS_SENSOR_COLS if c in gpvs_clean.columns]

GPVS_TARGET = 'fault_type'

# Encode fault_type labels (F0–F7 -> 0–7)
le = LabelEncoder()
gpvs_clean['fault_label'] = le.fit_transform(gpvs_clean[GPVS_TARGET])

print(f'Sensor features: {GPVS_SENSOR_COLS}')
print(f'Target: {GPVS_TARGET}')
print(f'Label encoding: {dict(zip(le.classes_, le.transform(le.classes_)))}')
print(f'\nClass distribution:')
print(gpvs_clean['fault_label'].value_counts().sort_index())
```

```
Sensor features: ['Ipv', 'Vpv', 'Vdc', 'ia', 'ib', 'ic', 'va', 'vb', 'vc',
                  'Iabc', 'If', 'Vabc', 'Vf']
Target:          fault_type
Label encoding:  {'F0': 0, 'F1': 1, 'F2': 2, 'F3': 3, 'F4': 4, 'F5': 5, 'F6': 6, 'F7': 7}
```

```
Class distribution:
fault_label
0      284729
1      268027
2      286143
3      173464
4      288028
5      287029
6      288030
7      288030
Name: count, dtype: int64
```

3.2 GPVS — Scale Features

```
In [27]: X_gpvs = gpvs_clean[GPVS_SENSOR_COLS]
         y_gpvs = gpvs_clean['fault_label']

         scaler_gpvs = StandardScaler()
         X_gpvs_scaled = pd.DataFrame(
             scaler_gpvs.fit_transform(X_gpvs),
             columns=GPVS_SENSOR_COLS
         )

         print(f'Scaled feature matrix: {X_gpvs_scaled.shape}')
```

Scaled feature matrix: (2163480, 13)

3.3 GPVS — Balance Classes with SMOTE

Class imbalance check from EDA showed uneven scenario sizes (F3M had only 69,967 rows vs 144,015 for others). SMOTE applied to balance before splitting.

```
In [28]: print('Class distribution BEFORE SMOTE:')
         print(y_gpvs.value_counts().sort_index())

         MAX_ROWS = 500_000
         if len(X_gpvs_scaled) > MAX_ROWS:
             print(f'\nDataset large ({len(X_gpvs_scaled):,} rows). Sampling {MAX_ROWS} rows')
             sample_idx = np.random.choice(len(X_gpvs_scaled), MAX_ROWS, replace=False)
             X_sample = X_gpvs_scaled.iloc[sample_idx]
             y_sample = y_gpvs.iloc[sample_idx]
         else:
             X_sample = X_gpvs_scaled
             y_sample = y_gpvs

         smote = SMOTE(random_state=42)
         X_gpvs_balanced, y_gpvs_balanced = smote.fit_resample(X_sample, y_sample)
```

```

print(f'\nClass distribution AFTER SMOTE:')
unique, counts = np.unique(y_gpvs_balanced, return_counts=True)
for u, c in zip(unique, counts):
    print(f' {le.inverse_transform([u])[0]} (class {u}): {c:,}')

# Plot before/after
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
pd.Series(y_sample.values).value_counts().sort_index().plot(
    kind='bar', ax=axes[0], color='steelblue', edgecolor='black', title='Before SMOTE')
pd.Series(y_gpvs_balanced).value_counts().sort_index().plot(
    kind='bar', ax=axes[1], color='mediumseagreen', edgecolor='black', title='After SMOTE')
for ax in axes:
    ax.set_xlabel('Fault Class')
    ax.set_ylabel('Count')
    ax.tick_params(axis='x', rotation=0)
plt.suptitle('GPVS - Class Balance Before vs After SMOTE', fontsize=13)
plt.tight_layout()
plt.savefig('/home/sagemaker-user/dataset1/gpvs_smote_balance.png', dpi=150)
plt.show()

```

Class distribution BEFORE SMOTE:

```

fault_label
0    284729
1    268027
2    286143
3    173464
4    288028
5    287029
6    288030
7    288030

```

Name: count, dtype: int64

Dataset large (2,163,480 rows). Sampling 500,000 rows for SMOTE...

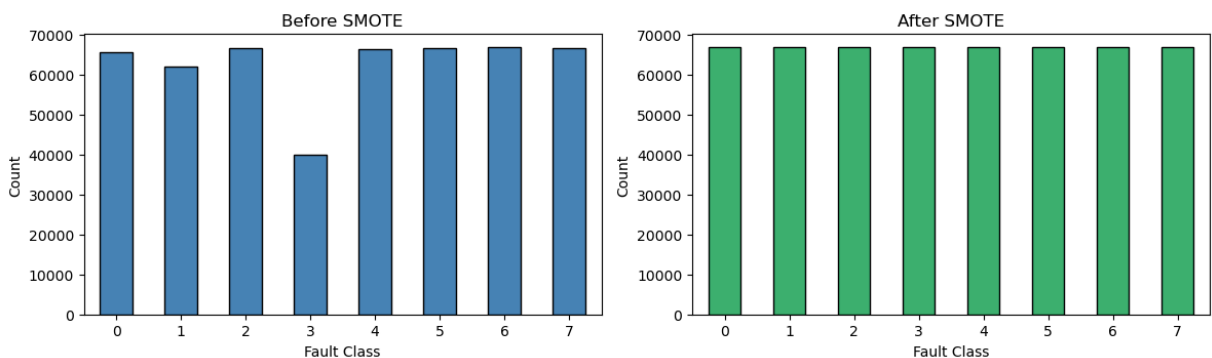
Class distribution AFTER SMOTE:

```

F0 (class 0): 66,712
F1 (class 1): 66,712
F2 (class 2): 66,712
F3 (class 3): 66,712
F4 (class 4): 66,712
F5 (class 5): 66,712
F6 (class 6): 66,712
F7 (class 7): 66,712

```

GPVS — Class Balance Before vs After SMOTE



3.4 GPVS — Train / Validation / Test Split (70/15/15)

```
In [29]: X_train_g, X_temp_g, y_train_g, y_temp_g = train_test_split(
        X_gpvs_balanced, y_gpvs_balanced,
        test_size=0.30, random_state=42, stratify=y_gpvs_balanced)
X_val_g, X_test_g, y_val_g, y_test_g = train_test_split(
        X_temp_g, y_temp_g,
        test_size=0.50, random_state=42, stratify=y_temp_g)

print('GPVS Split:')
print(f'  Train:      {len(X_train_g):,}')
print(f'  Validation: {len(X_val_g):,}')
print(f'  Test:       {len(X_test_g):,}')
```

```
GPVS Split:
  Train:      373,587
  Validation: 80,054
  Test:       80,055
```

4. Save Prepared Data to S3

All splits saved as headerless CSV under `s3://solaris-analytics-bucket/prepared/`

(SageMaker built-in algorithms expect target as first column, no header)

```
In [30]: prepared_dir = '/home/sagemaker-user/dataset1/prepared/'
os.makedirs(prepared_dir, exist_ok=True)

def save_and_upload(X, y, name, target_col):
    df = X.copy()
    y_vals = y.values if hasattr(y, 'values') else np.array(y)
    df.insert(0, target_col, y_vals) # target must be first column
    local_path = os.path.join(prepared_dir, f'{name}.csv')
    df.to_csv(local_path, index=False, header=False)
    s3_key = f'prepared/{name}.csv'
    s3.upload_file(local_path, BUCKET, s3_key)
    print(f'Uploaded {name}.csv ({len(df):,} rows) -> s3://{BUCKET}/{s3_key}')
    return f's3://{BUCKET}/{s3_key}'

# NSRDB irradiance splits
nsrdb_train_uri = save_and_upload(X_train_n, y_train_n, 'nsrdb_train', 'GHI')
nsrdb_val_uri   = save_and_upload(X_val_n,   y_val_n,   'nsrdb_val',   'GHI')
nsrdb_test_uri  = save_and_upload(X_test_n,  y_test_n,  'nsrdb_test',  'GHI')

# GPVS fault classification splits
gpvs_train_uri = save_and_upload(X_train_g, y_train_g, 'gpvs_train', 'fault')
gpvs_val_uri   = save_and_upload(X_val_g,   y_val_g,   'gpvs_val',   'fault')
gpvs_test_uri  = save_and_upload(X_test_g,  y_test_g,  'gpvs_test',  'fault')

print('\nAll prepared datasets uploaded to S3.')
```

```

Uploaded nsrdb_train.csv (122,591 rows) -> s3://solaris-analytics-bucket/
prepared/nsrdb_train.csv
Uploaded nsrdb_val.csv (26,270 rows) -> s3://solaris-analytics-bucket/pre
pared/nsrdb_val.csv
Uploaded nsrdb_test.csv (26,270 rows) -> s3://solaris-analytics-bucket/pr
epared/nsrdb_test.csv
Uploaded gpvs_train.csv (373,587 rows) -> s3://solaris-analytics-bucket/p
repared/gpvs_train.csv
Uploaded gpvs_val.csv (80,054 rows) -> s3://solaris-analytics-bucket/prep
ared/gpvs_val.csv
Uploaded gpvs_test.csv (80,055 rows) -> s3://solaris-analytics-bucket/pre
pared/gpvs_test.csv

```

All prepared datasets uploaded to S3.

5. Model Training — XGBoost for Fault Classification (GPVS)

Task: Classify fault type (F0–F7) from electrical sensor readings

Algorithm: SageMaker built-in XGBoost

Target metric: Weighted F1-score > 0.85

```

In [31]: num_classes = len(le.classes_)
xgb_image_uri = image_uris.retrieve('xgboost', REGION, version='1.7-1')
print(f'XGBoost image: {xgb_image_uri}')
print(f'Classes: {num_classes} ({list(le.classes_)})')

xgb_estimator = Estimator(
    image_uri = xgb_image_uri,
    role = role,
    instance_count = 1,
    instance_type = 'ml.m5.xlarge',
    output_path = f's3://{BUCKET}/models/xgboost/',
    sagemaker_session = session
)

xgb_estimator.set_hyperparameters(
    objective = 'multi:softmax',
    num_class = num_classes,
    num_round = 100,
    max_depth = 6,
    eta = 0.2,
    subsample = 0.8,
    colsample_bytree = 0.8,
    eval_metric = 'merror'
)

xgb_estimator.fit(
    inputs={
        'train': TrainingInput(gpvs_train_uri, content_type='text/csv')
        'validation': TrainingInput(gpvs_val_uri, content_type='text/csv')
    }
)

```

```
    },  
    wait=True  
)  
print('XGBoost training complete.')
```

INFO:sagemaker:Creating training-job with name: sagemaker-xgboost-2026-03-22-23-33-49-181

```

XGBoost image: 683313688378.dkr.ecr.us-east-1.amazonaws.com/sagemaker-xgboost:1.7-1
Classes: 8 (['F0', 'F1', 'F2', 'F3', 'F4', 'F5', 'F6', 'F7'])
2026-03-22 23:33:50 Starting - Starting the training job...
2026-03-22 23:34:05 Starting - Preparing the instances for training...
2026-03-22 23:34:51 Downloading - Downloading the training image.....
2026-03-22 23:35:42 Training - Training image download completed. Training in progress..
/miniconda3/lib/python3.9/site-packages/sagemaker_containers/_server.py:22: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
  import pkg_resources
[2026-03-22 23:35:53.022 ip-10-0-74-55.ec2.internal:7 INFO utils.py:28] RULE_JOB_STOP_SIGNAL_FILENAME: None
[2026-03-22 23:35:53.091 ip-10-0-74-55.ec2.internal:7 INFO profiler_config_parser.py:111] User has disabled profiler.
[2026-03-22:23:35:53:INFO] Imported framework sagemaker_xgboost_container.training
[2026-03-22:23:35:53:INFO] Failed to parse hyperparameter eval_metric value merror to Json.
Returning the value itself
[2026-03-22:23:35:53:INFO] Failed to parse hyperparameter objective value multi:softmax to Json.
Returning the value itself
[2026-03-22:23:35:53:INFO] No GPUs detected (normal if no gpus installed)
[2026-03-22:23:35:53:INFO] Running XGBoost Sagemaker in algorithm mode
[2026-03-22:23:35:53:INFO] Determined 0 GPU(s) available on the instance.
[2026-03-22:23:35:53:INFO] Determined delimiter of CSV input is ','
[2026-03-22:23:35:53:INFO] Determined delimiter of CSV input is ','
[2026-03-22:23:35:53:INFO] File path /opt/ml/input/data/train of input files
[2026-03-22:23:35:53:INFO] Making smlinks from folder /opt/ml/input/data/train to folder /tmp/sagemaker_xgboost_input_data
[2026-03-22:23:35:53:INFO] creating symlink between Path /opt/ml/input/data/train/gpvs_train.csv and destination /tmp/sagemaker_xgboost_input_data/gpvs_train.csv-8513690482331446650
[2026-03-22:23:35:53:INFO] files path: /tmp/sagemaker_xgboost_input_data
[2026-03-22:23:35:53:INFO] Determined delimiter of CSV input is ','
[2026-03-22:23:35:53:INFO] File path /opt/ml/input/data/validation of input files
[2026-03-22:23:35:53:INFO] Making smlinks from folder /opt/ml/input/data/validation to folder /tmp/sagemaker_xgboost_input_data
[2026-03-22:23:35:53:INFO] creating symlink between Path /opt/ml/input/data/validation/gpvs_val.csv and destination /tmp/sagemaker_xgboost_input_data/gpvs_val.csv-1954356500625305614
[2026-03-22:23:35:53:INFO] files path: /tmp/sagemaker_xgboost_input_data
[2026-03-22:23:35:53:INFO] Determined delimiter of CSV input is ','
[2026-03-22:23:35:53:INFO] Single node training.
[2026-03-22:23:35:53:INFO] Train matrix has 373587 rows and 13 columns
[2026-03-22:23:35:53:INFO] Validation matrix has 80054 rows
[2026-03-22 23:35:53.999 ip-10-0-74-55.ec2.internal:7 INFO json_config.py:92] Creating hook from json_config at /opt/ml/input/config/debughookconfig.json.
[2026-03-22 23:35:53.999 ip-10-0-74-55.ec2.internal:7 INFO hook.py:206] tensorboard_dir has not been set for the hook. SMDDebug will not be exporting tensorboard summaries.

```

```
[2026-03-22 23:35:54.000 ip-10-0-74-55.ec2.internal:7 INFO hook.py:259] Saving to /opt/ml/output/tensors
[2026-03-22 23:35:54.000 ip-10-0-74-55.ec2.internal:7 INFO state_store.py:77] The checkpoint config file /opt/ml/input/config/checkpointconfig.json does not exist.
[2026-03-22 23:35:54.000 ip-10-0-74-55.ec2.internal:7 INFO hook.py:427] Monitoring the collections: metrics
[2026-03-22 23:35:56.807 ip-10-0-74-55.ec2.internal:7 INFO hook.py:491] Hook is writing from the hook with pid: 7
[0]#011train-merror:0.23592#011validation-merror:0.23734
[1]#011train-merror:0.20357#011validation-merror:0.20807
[2]#011train-merror:0.18191#011validation-merror:0.18641
[3]#011train-merror:0.17125#011validation-merror:0.17528
[4]#011train-merror:0.16662#011validation-merror:0.17146
[5]#011train-merror:0.16438#011validation-merror:0.16872
[6]#011train-merror:0.15854#011validation-merror:0.16268
[7]#011train-merror:0.15466#011validation-merror:0.15887
[8]#011train-merror:0.14837#011validation-merror:0.15191
[9]#011train-merror:0.14691#011validation-merror:0.15054
[10]#011train-merror:0.14490#011validation-merror:0.14850
[11]#011train-merror:0.14350#011validation-merror:0.14689
[12]#011train-merror:0.14061#011validation-merror:0.14368
[13]#011train-merror:0.13809#011validation-merror:0.14144
[14]#011train-merror:0.13426#011validation-merror:0.13726
[15]#011train-merror:0.13258#011validation-merror:0.13585
[16]#011train-merror:0.12962#011validation-merror:0.13281
[17]#011train-merror:0.12789#011validation-merror:0.13122
[18]#011train-merror:0.12594#011validation-merror:0.12930
[19]#011train-merror:0.12411#011validation-merror:0.12738
[20]#011train-merror:0.12259#011validation-merror:0.12624
[21]#011train-merror:0.12129#011validation-merror:0.12538
[22]#011train-merror:0.11986#011validation-merror:0.12380
[23]#011train-merror:0.11745#011validation-merror:0.12111
[24]#011train-merror:0.11556#011validation-merror:0.11943
[25]#011train-merror:0.11204#011validation-merror:0.11633
[26]#011train-merror:0.11074#011validation-merror:0.11497
[27]#011train-merror:0.10981#011validation-merror:0.11367
[28]#011train-merror:0.10821#011validation-merror:0.11182
[29]#011train-merror:0.10628#011validation-merror:0.11020
[30]#011train-merror:0.10489#011validation-merror:0.10901
[31]#011train-merror:0.10199#011validation-merror:0.10648
[32]#011train-merror:0.10011#011validation-merror:0.10455
[33]#011train-merror:0.09788#011validation-merror:0.10217
[34]#011train-merror:0.09634#011validation-merror:0.10066
[35]#011train-merror:0.09515#011validation-merror:0.09936
[36]#011train-merror:0.09338#011validation-merror:0.09801
[37]#011train-merror:0.09170#011validation-merror:0.09626
[38]#011train-merror:0.08964#011validation-merror:0.09406
[39]#011train-merror:0.08776#011validation-merror:0.09221
[40]#011train-merror:0.08510#011validation-merror:0.08965
[41]#011train-merror:0.08376#011validation-merror:0.08822
[42]#011train-merror:0.08182#011validation-merror:0.08677
[43]#011train-merror:0.07952#011validation-merror:0.08446
[44]#011train-merror:0.07855#011validation-merror:0.08352
[45]#011train-merror:0.07716#011validation-merror:0.08216
```

[46]#011train-merror:0.07516#011validation-merror:0.08003
[47]#011train-merror:0.07381#011validation-merror:0.07876
[48]#011train-merror:0.07148#011validation-merror:0.07662
[49]#011train-merror:0.07058#011validation-merror:0.07542
[50]#011train-merror:0.06899#011validation-merror:0.07401
[51]#011train-merror:0.06758#011validation-merror:0.07275
[52]#011train-merror:0.06656#011validation-merror:0.07165
[53]#011train-merror:0.06465#011validation-merror:0.06987
[54]#011train-merror:0.06343#011validation-merror:0.06860
[55]#011train-merror:0.06275#011validation-merror:0.06783
[56]#011train-merror:0.06202#011validation-merror:0.06728
[57]#011train-merror:0.06138#011validation-merror:0.06669
[58]#011train-merror:0.05975#011validation-merror:0.06542
[59]#011train-merror:0.05890#011validation-merror:0.06439
[60]#011train-merror:0.05827#011validation-merror:0.06392
[61]#011train-merror:0.05728#011validation-merror:0.06296
[62]#011train-merror:0.05661#011validation-merror:0.06221
[63]#011train-merror:0.05592#011validation-merror:0.06167
[64]#011train-merror:0.05486#011validation-merror:0.06070
[65]#011train-merror:0.05369#011validation-merror:0.05977
[66]#011train-merror:0.05291#011validation-merror:0.05855
[67]#011train-merror:0.05196#011validation-merror:0.05751
[68]#011train-merror:0.05112#011validation-merror:0.05685
[69]#011train-merror:0.05042#011validation-merror:0.05607
[70]#011train-merror:0.04991#011validation-merror:0.05577
[71]#011train-merror:0.04816#011validation-merror:0.05411
[72]#011train-merror:0.04749#011validation-merror:0.05346
[73]#011train-merror:0.04675#011validation-merror:0.05290
[74]#011train-merror:0.04519#011validation-merror:0.05145
[75]#011train-merror:0.04442#011validation-merror:0.05054
[76]#011train-merror:0.04372#011validation-merror:0.04993
[77]#011train-merror:0.04319#011validation-merror:0.04917
[78]#011train-merror:0.04267#011validation-merror:0.04864
[79]#011train-merror:0.04203#011validation-merror:0.04815
[80]#011train-merror:0.04167#011validation-merror:0.04793
[81]#011train-merror:0.04118#011validation-merror:0.04751
[82]#011train-merror:0.04061#011validation-merror:0.04691
[83]#011train-merror:0.03993#011validation-merror:0.04637
[84]#011train-merror:0.03938#011validation-merror:0.04593
[85]#011train-merror:0.03873#011validation-merror:0.04546
[86]#011train-merror:0.03795#011validation-merror:0.04463
[87]#011train-merror:0.03733#011validation-merror:0.04422
[88]#011train-merror:0.03694#011validation-merror:0.04392
[89]#011train-merror:0.03649#011validation-merror:0.04337
[90]#011train-merror:0.03620#011validation-merror:0.04293
[91]#011train-merror:0.03550#011validation-merror:0.04237
[92]#011train-merror:0.03522#011validation-merror:0.04230
[93]#011train-merror:0.03468#011validation-merror:0.04206
[94]#011train-merror:0.03446#011validation-merror:0.04172
[95]#011train-merror:0.03413#011validation-merror:0.04137
[96]#011train-merror:0.03370#011validation-merror:0.04118
[97]#011train-merror:0.03334#011validation-merror:0.04082

2026-03-22 23:39:48 Uploading - Uploading generated training model[98]#011train-merror:0.03289#011validation-merror:0.04030
[99]#011train-merror:0.03244#011validation-merror:0.03994

2026-03-22 23:40:01 Completed - Training job completed
 Training seconds: 335
 Billable seconds: 335
 XGBoost training complete.

5.1 XGBoost — Evaluate on Test Set

```
In [32]: xgb_predictor = xgb_estimator.deploy(
    initial_instance_count=1,
    instance_type='ml.m5.xlarge',
    serializer=CSVSerializer()
)

def predict_batches(predictor, X, batch_size=500):
    preds = []
    for i in range(0, len(X), batch_size):
        batch = X.iloc[i:i+batch_size] if hasattr(X, 'iloc') else X[i:i+bat
        result = predictor.predict(batch.values if hasattr(batch, 'values')
        preds.extend([float(x) for x in result.decode('utf-8').strip().split
    return np.array(preds)

y_pred_g = predict_batches(xgb_predictor, X_test_g)
f1 = f1_score(y_test_g, y_pred_g, average='weighted')

print('=== XGBoost Fault Classification Results ===')
print(f'Weighted F1-Score: {f1:.4f} (target > 0.85)')
print(f'Target Met: {"YES ✓" if f1 > 0.85 else "NO – consider tuning hyperpa
print(f'\nClassification Report:')
print(classification_report(y_test_g, y_pred_g, target_names=le.classes_))

fig, ax = plt.subplots(figsize=(9, 7))
cm = confusion_matrix(y_test_g, y_pred_g)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', ax=ax,
            xticklabels=le.classes_, yticklabels=le.classes_)
ax.set_title(f'XGBoost – Confusion Matrix (F1={f1:.3f})', fontsize=13)
ax.set_xlabel('Predicted')
ax.set_ylabel('Actual')
plt.tight_layout()
plt.savefig('/home/sagemaker-user/dataset1/xgb_confusion_matrix.png', dpi=15
plt.show()

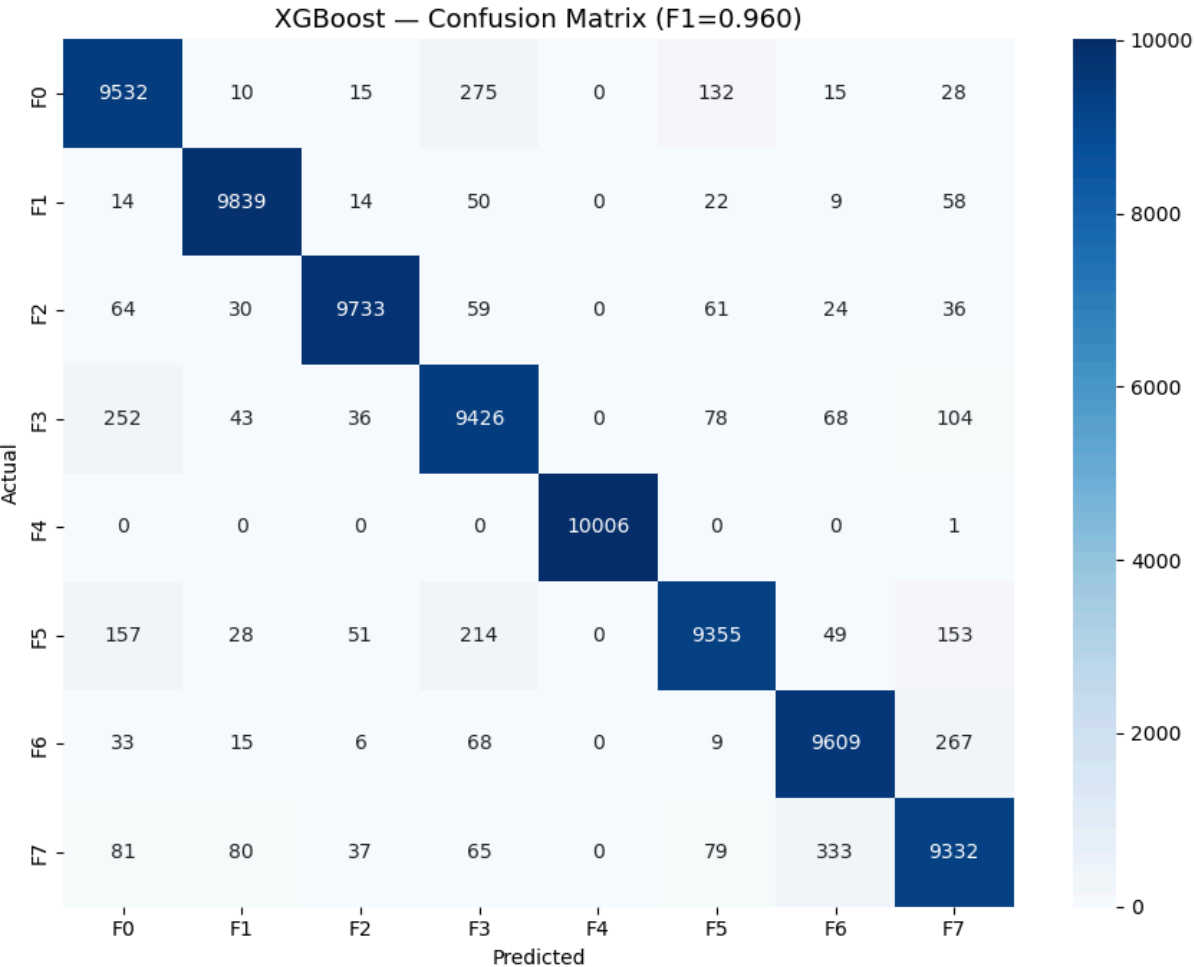
xgb_predictor.delete_endpoint()
print('Endpoint deleted.')
```

```
INFO:sagemaker:Creating model with name: sagemaker-xgboost-2026-03-22-23-40-
40-565
INFO:sagemaker:Creating endpoint-config with name sagemaker-xgboost-2026-03-
22-23-40-40-565
INFO:sagemaker:Creating endpoint with name sagemaker-xgboost-2026-03-22-23-4
0-40-565
```

-----!==== XGBoost Fault Classification Results ===
Weighted F1-Score: 0.9598 (target > 0.85)
Target Met: YES ✓

Classification Report:

	precision	recall	f1-score	support
F0	0.94	0.95	0.95	10007
F1	0.98	0.98	0.98	10006
F2	0.98	0.97	0.98	10007
F3	0.93	0.94	0.93	10007
F4	1.00	1.00	1.00	10007
F5	0.96	0.93	0.95	10007
F6	0.95	0.96	0.96	10007
F7	0.94	0.93	0.93	10007
accuracy			0.96	80055
macro avg	0.96	0.96	0.96	80055
weighted avg	0.96	0.96	0.96	80055



INFO:sagemaker:Deleting endpoint configuration with name: sagemaker-xgboost-2026-03-22-23-40-40-565
INFO:sagemaker:Deleting endpoint with name: sagemaker-xgboost-2026-03-22-23-40-40-565
Endpoint deleted.

6. Model Training — Linear Learner for Irradiance Forecasting (NSRDB)

Task: Predict GHI solar irradiance from weather + temporal features

Algorithm: SageMaker built-in Linear Learner (regression)

Target metric: 20% RMSE reduction over historical average baseline

```
In [33]: baseline_pred = np.full(len(y_test_n), y_train_n.mean())
baseline_rmse = np.sqrt(mean_squared_error(y_test_n, baseline_pred))
target_rmse   = baseline_rmse * 0.80

print(f'Baseline RMSE (historical mean): {baseline_rmse:.4f} W/m2')
print(f'Target RMSE (20% reduction):    {target_rmse:.4f} W/m2')
```

Baseline RMSE (historical mean): 290.4492 W/m2

Target RMSE (20% reduction): 232.3594 W/m2

```
In [34]: ll_image_uri = image_uris.retrieve('linear-learner', REGION)
print(f'Linear Learner image: {ll_image_uri}')

ll_estimator = Estimator(
    image_uri      = ll_image_uri,
    role           = role,
    instance_count = 1,
    instance_type  = 'ml.m5.xlarge',
    output_path    = f's3://{BUCKET}/models/linear-learner/',
    sagemaker_session = session
)

ll_estimator.set_hyperparameters(
    predictor_type = 'regressor',
    mini_batch_size = 200,
    epochs         = 15,
    learning_rate  = 'auto',
    loss           = 'squared_loss',
    normalize_data = True
)

ll_estimator.fit(
    inputs={
        'train': TrainingInput(nsrdn_train_uri, content_type='text/csv'),
        'validation': TrainingInput(nsrdn_val_uri, content_type='text/csv')
    },
    wait=True
)
print('Linear Learner training complete.')
```

```
INFO:sagemaker.image_uris:Same images used for training and inference. Defaulting to image scope: inference.  
INFO:sagemaker.image_uris:Ignoring unnecessary instance type: None.  
INFO:sagemaker.telemetry.telemetry_logging:SageMaker Python SDK will collect telemetry to help us better understand our user's needs, diagnose issues, and deliver additional features.  
To opt out of telemetry, please disable via TelemetryOptOut parameter in SDK defaults config. For more information, refer to https://sagemaker.readthedocs.io/en/stable/overview.html#configuring-and-using-defaults-with-the-sagemaker-python-sdk.  
INFO:sagemaker:Creating training-job with name: linear-learner-2026-03-22-23-44-21-034
```

```

Linear Learner image: 382416733822.dkr.ecr.us-east-1.amazonaws.com/linear-le
arner:1
2026-03-22 23:44:22 Starting - Starting the training job...
2026-03-22 23:44:37 Starting - Preparing the instances for training...
2026-03-22 23:45:23 Downloading - Downloading the training image.....
2026-03-22 23:46:24 Training - Training image download completed. Training i
n progress...
Docker entrypoint called with argument(s): train
Running default environment configuration script
[03/22/2026 23:46:35 INFO 139950999783232] Reading default configuration fro
m /opt/amazon/lib/python3.8/site-packages/algorithm/resources/default-input.
json: {'mini_batch_size': '1000', 'epochs': '15', 'feature_dim': 'auto', 'us
e_bias': 'true', 'binary_classifier_model_selection_criteria': 'accuracy',
'f_beta': '1.0', 'target_recall': '0.8', 'target_precision': '0.8', 'num_mod
els': 'auto', 'num_calibration_samples': '10000000', 'init_method': 'unifor
m', 'init_scale': '0.07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimiz
er': 'auto', 'loss': 'auto', 'margin': '1.0', 'quantile': '0.5', 'loss_insen
sitivity': '0.01', 'huber_delta': '1.0', 'num_classes': '1', 'accuracy_top_
k': '3', 'wd': 'auto', 'l1': 'auto', 'momentum': 'auto', 'learning_rate': 'a
uto', 'beta_1': 'auto', 'beta_2': 'auto', 'bias_lr_mult': 'auto', 'bias_wd_m
ult': 'auto', 'use_lr_scheduler': 'true', 'lr_scheduler_step': 'auto', 'lr_s
cheduler_factor': 'auto', 'lr_scheduler_minimum_lr': 'auto', 'positive_examp
le_weight_mult': '1.0', 'balance_multiclass_weights': 'false', 'normalize_da
ta': 'true', 'normalize_label': 'auto', 'unbias_data': 'auto', 'unbias_labe
l': 'auto', 'num_point_for_scaler': '10000', '_kvstore': 'auto', '_num_gpu
s': 'auto', '_num_kv_servers': 'auto', '_log_level': 'info', '_tuning_object
ive_metric': '', 'early_stopping_patience': '3', 'early_stopping_tolerance':
'0.001', '_enable_profiler': 'false'}
[03/22/2026 23:46:35 INFO 139950999783232] Merging with provided configurati
on from /opt/ml/input/config/hyperparameters.json: {'epochs': '15', 'learnin
g_rate': 'auto', 'loss': 'squared_loss', 'mini_batch_size': '200', 'normaliz
e_data': 'True', 'predictor_type': 'regressor'}
[03/22/2026 23:46:35 INFO 139950999783232] Final configuration: {'mini_batch
_size': '200', 'epochs': '15', 'feature_dim': 'auto', 'use_bias': 'true', 'b
inary_classifier_model_selection_criteria': 'accuracy', 'f_beta': '1.0', 'ta
rget_recall': '0.8', 'target_precision': '0.8', 'num_models': 'auto', 'num_c
alibration_samples': '10000000', 'init_method': 'uniform', 'init_scale': '0.
07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimizer': 'auto', 'loss':
'squared_loss', 'margin': '1.0', 'quantile': '0.5', 'loss_insensitivity':
'0.01', 'huber_delta': '1.0', 'num_classes': '1', 'accuracy_top_k': '3', 'w
d': 'auto', 'l1': 'auto', 'momentum': 'auto', 'learning_rate': 'auto', 'beta
_1': 'auto', 'beta_2': 'auto', 'bias_lr_mult': 'auto', 'bias_wd_mult': 'aut
o', 'use_lr_scheduler': 'true', 'lr_scheduler_step': 'auto', 'lr_scheduler_f
actor': 'auto', 'lr_scheduler_minimum_lr': 'auto', 'positive_example_weight_
mult': '1.0', 'balance_multiclass_weights': 'false', 'normalize_data': 'Tru
e', 'normalize_label': 'auto', 'unbias_data': 'auto', 'unbias_label': 'aut
o', 'num_point_for_scaler': '10000', '_kvstore': 'auto', '_num_gpus': 'aut
o', '_num_kv_servers': 'auto', '_log_level': 'info', '_tuning_objective_metr
ic': '', 'early_stopping_patience': '3', 'early_stopping_tolerance': '0.00
1', '_enable_profiler': 'false', 'predictor_type': 'regressor'}
/opt/amazon/lib/python3.8/site-packages/mxnet/model.py:97: SyntaxWarning: "i
s" with a literal. Did you mean "=="?
    if num_device is 1 and 'dist' not in kvstore:
/opt/amazon/lib/python3.8/site-packages/scipy/optimize/_shgo.py:495: SyntaxW
arning: "is" with a literal. Did you mean "=="?
    if cons['type'] is 'ineq':
/opt/amazon/lib/python3.8/site-packages/scipy/optimize/_shgo.py:743: SyntaxW

```

```

arning: "is not" with a literal. Did you mean "!="?
    if len(self.X_min) is not 0:
[03/22/2026 23:46:37 WARNING 139950999783232] Loggers have already been setu
p.
[03/22/2026 23:46:37 INFO 139950999783232] Final configuration: {'mini_batch
_size': '200', 'epochs': '15', 'feature_dim': 'auto', 'use_bias': 'true', 'b
inary_classifier_model_selection_criteria': 'accuracy', 'f_beta': '1.0', 'ta
rget_recall': '0.8', 'target_precision': '0.8', 'num_models': 'auto', 'num_c
alibration_samples': '1000000', 'init_method': 'uniform', 'init_scale': '0.
07', 'init_sigma': '0.01', 'init_bias': '0.0', 'optimizer': 'auto', 'loss':
'squared_loss', 'margin': '1.0', 'quantile': '0.5', 'loss_insensitivity':
'0.01', 'huber_delta': '1.0', 'num_classes': '1', 'accuracy_top_k': '3', 'w
d': 'auto', 'l1': 'auto', 'momentum': 'auto', 'learning_rate': 'auto', 'beta
_1': 'auto', 'beta_2': 'auto', 'bias_lr_mult': 'auto', 'bias_wd_mult': 'aut
o', 'use_lr_scheduler': 'true', 'lr_scheduler_step': 'auto', 'lr_scheduler_f
actor': 'auto', 'lr_scheduler_minimum_lr': 'auto', 'positive_example_weight_
mult': '1.0', 'balance_multiclass_weights': 'false', 'normalize_data': 'Tru
e', 'normalize_label': 'auto', 'unbias_data': 'auto', 'unbias_label': 'aut
o', 'num_point_for_scaler': '10000', '_kvstore': 'auto', '_num_gpus': 'aut
o', '_num_kv_servers': 'auto', '_log_level': 'info', '_tuning_objective_metr
ic': '', 'early_stopping_patience': '3', 'early_stopping_tolerance': '0.00
1', '_enable_profiler': 'false', 'predictor_type': 'regressor'}
[03/22/2026 23:46:37 WARNING 139950999783232] Loggers have already been setu
p.
Process 7 is a worker.
[03/22/2026 23:46:37 INFO 139950999783232] Using default worker.
[03/22/2026 23:46:37 INFO 139950999783232] Checkpoint loading and saving are
disabled.
[03/22/2026 23:46:37 INFO 139950999783232] Create Store: local
[03/22/2026 23:46:38 INFO 139950999783232] Scaler algorithm parameters
<algorithm.scaler.ScalerAlgorithmStable object at 0x7f48515115e0>
[03/22/2026 23:46:38 INFO 139950999783232] Scaling model computed with param
eters:
    {'stdev_label':
[289.43982]
<NDArray 1 @cpu(0)>, 'stdev_weight':
[1.0030912  0.9847499  0.99496585 1.0099543  1.0000852  1.0010041
 0.994852  0.99423265 1.0065767 ]
<NDArray 9 @cpu(0)>, 'mean_label':
[409.7331]
<NDArray 1 @cpu(0)>, 'mean_weight':
[ 0.02480434 -0.00467327 -0.00497079 -0.00833221  0.01480358  0.00905399
 -0.00566883 -0.01502212 -0.00654342]
<NDArray 9 @cpu(0)>}]
/opt/amazon/python3.8/lib/python3.8/subprocess.py:848: RuntimeWarning: line
buffering (buffering=1) isn't supported in binary mode, the default buffer s
ize will be used
    self.stdout = io.open(c2pread, 'rb', bufsize)
[03/22/2026 23:46:38 INFO 139950999783232] nvidia-smi: took 0.032 seconds to
run.
[03/22/2026 23:46:38 INFO 139950999783232] nvidia-smi identified 0 GPUs.
[03/22/2026 23:46:38 INFO 139950999783232] Number of GPUs being used: 0
#metrics {"StartTime": 1774223198.2810094, "EndTime": 1774223198.2810435, "D
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[03/22/2026 23:46:48 INFO 139950999783232] #quality_metric: host=algo-1, epo
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[03/22/2026 23:46:50 INFO 139950999783232] #quality_metric: host=algo-1, epo
ch=0, validation mse_objective <loss>=6016.619856299962
[03/22/2026 23:46:50 INFO 139950999783232] #early_stopping_criteria_metric:
host=algo-1, epoch=0, criteria=mse_objective, value=6016.619856299962
[03/22/2026 23:46:50 INFO 139950999783232] Epoch 0: Loss improved. Updating
best model
[03/22/2026 23:46:50 INFO 139950999783232] Saving model for epoch: 0

```

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[03/22/2026 23:46:50 INFO 139950999783232] Saved checkpoint to "/tmp/tmp5mmq
w0jv/mx-mod-0000.params"
[03/22/2026 23:46:50 INFO 139950999783232] #progress_metric: host=algo-1, co
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[03/22/2026 23:47:00 INFO 139950999783232] #quality_metric: host=algo-1, epo  
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ch=1, validation mse_objective <loss>=6003.574281499809  
[03/22/2026 23:47:01 INFO 139950999783232] #early_stopping_criteria_metric:  
host=algo-1, epoch=1, criteria=mse_objective, value=6003.574281499809  
[03/22/2026 23:47:01 INFO 139950999783232] Epoch 1: Loss improved. Updating  
best model  
[03/22/2026 23:47:01 INFO 139950999783232] Saving model for epoch: 1  
[03/22/2026 23:47:01 INFO 139950999783232] Saved checkpoint to "/tmp/tmpc3gb  
6lqd/mx-mod-0000.params"  
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[03/22/2026 23:47:11 INFO 139950999783232] #quality_metric: host=algo-1, epoch=2, train mse_objective <loss>=0.07076384903558719
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[03/22/2026 23:47:12 INFO 139950999783232] #quality_metric: host=algo-1, epo
ch=2, validation mse_objective <loss>=6001.297283022459
[03/22/2026 23:47:12 INFO 139950999783232] #early_stopping_criteria_metric:
host=algo-1, epoch=2, criteria=mse_objective, value=6001.262726018272
[03/22/2026 23:47:12 INFO 139950999783232] Saving model for epoch: 2
[03/22/2026 23:47:12 INFO 139950999783232] Saved checkpoint to "/tmp/tmp0vcg
sv9o/mx-mod-0000.params"
[03/22/2026 23:47:12 INFO 139950999783232] #progress_metric: host=algo-1, co
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[03/22/2026 23:47:23 INFO 139950999783232] #quality_metric: host=algo-1, epoch=3, validation mse_objective <loss>=6001.119915778455
[03/22/2026 23:47:23 INFO 139950999783232] #early_stopping_criteria_metric: host=algo-1, epoch=3, criteria=mse_objective, value=6001.1124500380665
[03/22/2026 23:47:23 INFO 139950999783232] Saving model for epoch: 3
[03/22/2026 23:47:23 INFO 139950999783232] Saved checkpoint to "/tmp/tmpiko9z8xl/mx-mod-0000.params"
[03/22/2026 23:47:23 INFO 139950999783232] #progress_metric: host=algo-1, completed 26.666666666666668 % of epochs
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[03/22/2026 23:47:33 INFO 139950999783232] #quality_metric: host=algo-1, epoch=4, train mse_objective <loss>=0.0706680094884112  
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[03/22/2026 23:47:34 INFO 139950999783232] #quality_metric: host=algo-1, epo
ch=4, validation mse_objective <loss>=6001.636283070042
[03/22/2026 23:47:34 INFO 139950999783232] #early_stopping_criteria_metric:
host=algo-1, epoch=4, criteria=mse_objective, value=6001.633601779597
[03/22/2026 23:47:34 INFO 139950999783232] Saving model for epoch: 4
[03/22/2026 23:47:34 INFO 139950999783232] Saved checkpoint to "/tmp/tmpszdx
uyt6/mx-mod-0000.params"
[03/22/2026 23:47:34 INFO 139950999783232] #progress_metric: host=algo-1, co
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[03/22/2026 23:47:34 INFO 139950999783232] #throughput_metric: host=algo-1,
train throughput=10918.26757951787 records/second

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2026-03-22 23:47:50 Uploading - Uploading generated training model#metrics
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```


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```

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[03/22/2026 23:47:45 INFO 139950999783232] #quality_metric: host=algo-1, epo
ch=5, validation mse_objective <loss>=6000.099557480015
[03/22/2026 23:47:45 INFO 139950999783232] #early_stopping_criteria_metric:
host=algo-1, epoch=5, criteria=mse_objective, value=6000.099557480015
[03/22/2026 23:47:45 INFO 139950999783232] Saving model for epoch: 5
[03/22/2026 23:47:45 INFO 139950999783232] Saved checkpoint to "/tmp/tmp46
x80x/mx-mod-0000.params"
[03/22/2026 23:47:45 INFO 139950999783232] Early stop condition met. Stoppin
g training.
[03/22/2026 23:47:45 INFO 139950999783232] #progress_metric: host=algo-1, co
mpleted 100 % epochs
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```

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91, "max": 122591}, "Max Batches Seen Between Resets": {"sum": 613.0, "count": 1, "min": 613, "max": 613}, "Reset Count": {"sum": 8.0, "count": 1, "min": 8, "max": 8}, "Number of Records Since Last Reset": {"sum": 122591.0, "count": 1, "min": 122591, "max": 122591}, "Number of Batches Since Last Reset": {"sum": 613.0, "count": 1, "min": 613, "max": 613}}
[03/22/2026 23:47:45 INFO 139950999783232] #throughput_metric: host=algo-1, train throughput=10935.58170274926 records/second
[03/22/2026 23:47:45 WARNING 139950999783232] wait_for_all_workers will not sync workers since the kv store is not running distributed
[03/22/2026 23:47:45 WARNING 139950999783232] wait_for_all_workers will not sync workers since the kv store is not running distributed
[03/22/2026 23:47:47 INFO 139950999783232] #early_stopping_criteria_metric: host=algo-1, epoch=5, criteria=mse_objective, value=6000.099557480015
[03/22/2026 23:47:47 INFO 139950999783232] #validation_score (algo-1) : ('mse_objective', 6003.574281499809)
[03/22/2026 23:47:47 INFO 139950999783232] #validation_score (algo-1) : ('mse', 6003.574281499809)
[03/22/2026 23:47:47 INFO 139950999783232] #validation_score (algo-1) : ('absolute_loss', 62.940420412572564)
[03/22/2026 23:47:47 INFO 139950999783232] #validation_score (algo-1) : ('rmse', 77.48273537698452)
[03/22/2026 23:47:47 INFO 139950999783232] #validation_score (algo-1) : ('r2', 0.9280900461820392)
[03/22/2026 23:47:47 INFO 139950999783232] #validation_score (algo-1) : ('mae', 62.940420328350086)
[03/22/2026 23:47:47 INFO 139950999783232] #quality_metric: host=algo-1, validation mse_objective <loss>=6003.574281499809
[03/22/2026 23:47:47 INFO 139950999783232] #quality_metric: host=algo-1, validation mse <loss>=6003.574281499809
[03/22/2026 23:47:47 INFO 139950999783232] #quality_metric: host=algo-1, validation absolute_loss <loss>=62.940420412572564
[03/22/2026 23:47:47 INFO 139950999783232] #quality_metric: host=algo-1, validation rmse <loss>=77.48273537698452
[03/22/2026 23:47:47 INFO 139950999783232] #quality_metric: host=algo-1, validation r2 <loss>=0.9280900461820392
[03/22/2026 23:47:47 INFO 139950999783232] #quality_metric: host=algo-1, validation mae <loss>=62.940420328350086
[03/22/2026 23:47:47 INFO 139950999783232] Best model found for hyperparameters: {"optimizer": "adam", "learning_rate": 0.005, "l1": 0.0, "wd": 0.0001, "lr_scheduler_step": 10, "lr_scheduler_factor": 0.99, "lr_scheduler_minimum_lr": 1e-05}
[03/22/2026 23:47:47 INFO 139950999783232] Saved checkpoint to "/tmp/tmpzgjo3qxz/mx-mod-0000.params"
[03/22/2026 23:47:47 INFO 139950999783232] Test data is not provided.
#metrics {"StartTime": 1774223197.8058407, "EndTime": 1774223267.615863, "Dimensions": {"Algorithm": "Linear Learner", "Host": "algo-1", "Operation": "training"}, "Metrics": {"initialize.time": {"sum": 463.0467891693115, "count": 1, "min": 463.0467891693115, "max": 463.0467891693115}, "epochs": {"sum": 15.0, "count": 1, "min": 15, "max": 15}, "check_early_stopping.time": {"sum": 3.0412673950195312, "count": 7, "min": 0.18525123596191406, "max": 1.0657310485839844}, "update.time": {"sum": 67437.7634525299, "count": 6, "min": 10620.591878890991, "max": 12139.956712722778}, "finalize.time": {"sum": 1875.537633895874, "count": 1, "min": 1875.537633895874, "max": 1875.537633895874}, "setuptime": {"sum": 2.0630359649658203, "count": 1, "min": 2.0630359649658203, "max": 2.0630359649658203}, "totaltime": {"sum": 69911.89670562744, "count": 1, "min": 69911.89670562744, "max": 69911.89670562744}}}

```

2026-03-22 23:48:08 Completed - Training job completed
 Training seconds: 190
 Billable seconds: 190
 Linear Learner training complete.

6.1 Linear Learner — Evaluate on Test Set

```
In [35]: ll_predictor = ll_estimator.deploy(
    initial_instance_count=1,
    instance_type='ml.m5.xlarge',
    serializer=CSVSerializer()
)

def predict_regression_batches(predictor, X, batch_size=500):
    preds = []
    for i in range(0, len(X), batch_size):
        batch = X.iloc[i:i+batch_size]
        result = predictor.predict(batch.values)
        parsed = json.loads(result)
        preds.extend([p['score'] for p in parsed['predictions']])
    return np.array(preds)

y_pred_n = predict_regression_batches(ll_predictor, X_test_n)
model_rmse = np.sqrt(mean_squared_error(y_test_n, y_pred_n))
rmse_pct = (baseline_rmse - model_rmse) / baseline_rmse * 100

print('=== Linear Learner Irradiance Forecasting Results ===')
print(f'Baseline RMSE: {baseline_rmse:.4f} W/m2')
print(f'Model RMSE: {model_rmse:.4f} W/m2')
print(f'RMSE Reduction: {rmse_pct:.1f}% (target >= 20%)')
print(f'Target Met: {"YES ✓" if rmse_pct >= 20 else "NO - consider XGBoost r

fig, ax = plt.subplots(figsize=(7, 6))
sample = min(2000, len(y_test_n))
ax.scatter(y_test_n.values[:sample], y_pred_n[:sample], alpha=0.3, s=8, color='blue')
ax.plot([y_test_n.min(), y_test_n.max()],
        [y_test_n.min(), y_test_n.max()], 'r--', lw=2, label='Perfect fit')
ax.set_xlabel('Actual GHI (W/m2)')
ax.set_ylabel('Predicted GHI (W/m2)')
ax.set_title(f'Linear Learner - Actual vs Predicted GHI\nRMSE={model_rmse:.2f}')
ax.legend()
plt.tight_layout()
plt.savefig('/home/sagemaker-user/dataset1/ll_actual_vs_predicted.png', dpi=300)
plt.show()

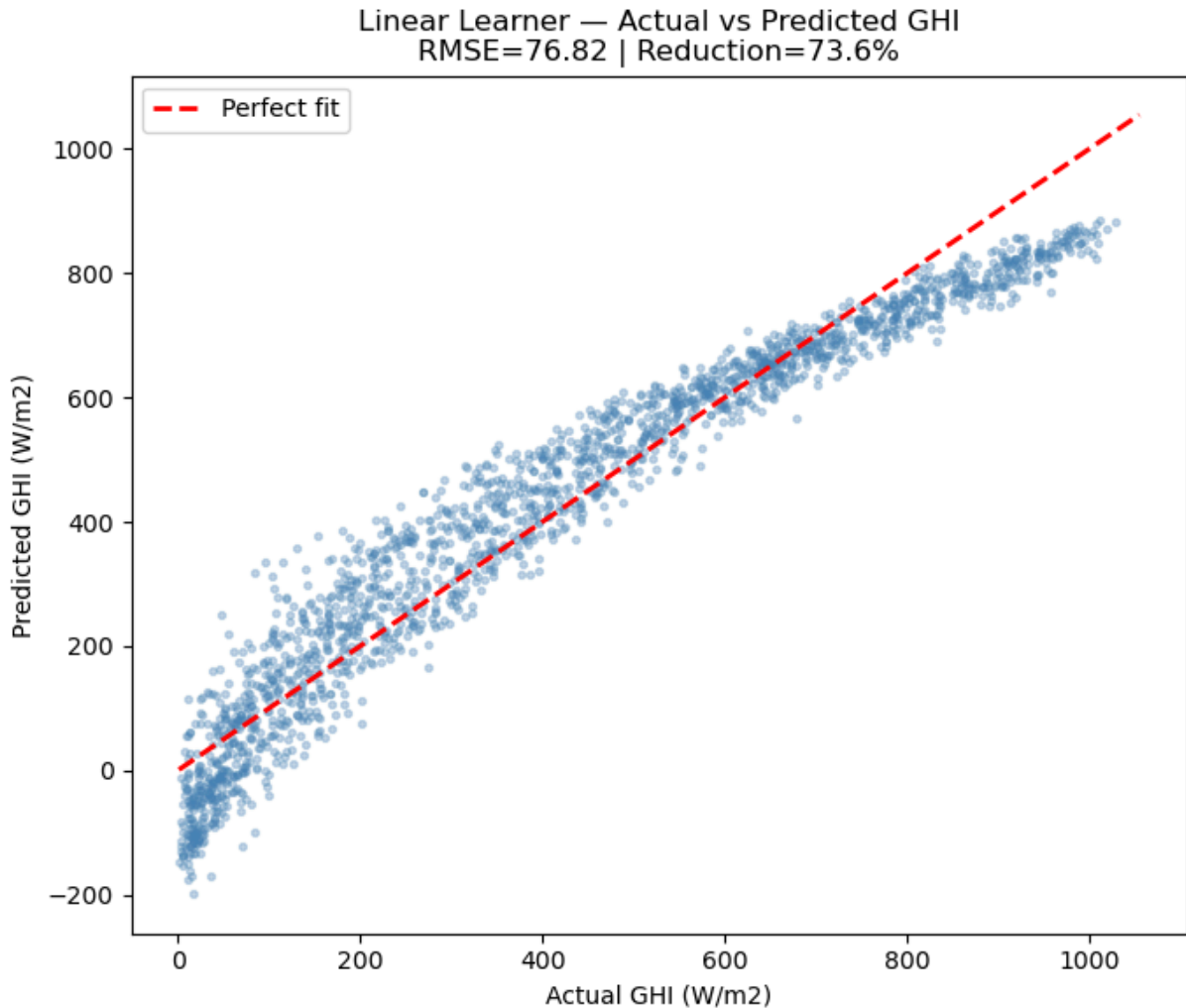
ll_predictor.delete_endpoint()
print('Endpoint deleted.')
```

```
INFO:sagemaker:Creating model with name: linear-learner-2026-03-22-23-48-40-010
INFO:sagemaker:Creating endpoint-config with name linear-learner-2026-03-22-23-48-40-010
INFO:sagemaker:Creating endpoint with name linear-learner-2026-03-22-23-48-40-010
```

```

-----!=== Linear Learner Irradiance Forecasting Results ===
Baseline RMSE: 290.4492 W/m2
Model RMSE:      76.8173 W/m2
RMSE Reduction: 73.6% (target >= 20%)
Target Met: YES ✓

```



```

INFO:sagemaker:Deleting endpoint configuration with name: linear-learner-2026-03-22-23-48-40-010
INFO:sagemaker:Deleting endpoint with name: linear-learner-2026-03-22-23-48-40-010
Endpoint deleted.

```

7. Final Summary

```

In [36]: print('=' * 62)
          SOLARIS ANALYTICS — DATA PREP & TRAINING SUMMARY
          print('=' * 62)

          print('\n[DATA PREPARATION — NSRDB]')
          print(f'  Raw rows:           {len(nsrdb_df):,}')
          print(f'  Cleaned (daytime):       {len(nsrdb_daytime):,}')
          print(f'  Features:               {NSRDB_FEATURES}')
          print(f'  Target:                 GHI (continuous)')

```



```
print(f'  Scaling:                StandardScaler')
print(f'  Split:                  70/15/15')
print(f'  Train: {len(X_train_n):,} | Val: {len(X_val_n):,} | Test: {len(X_t

print('\n[DATA PREPARATION - GPVS]')
print(f'  Raw rows:                    {len(gpvs_df):,}')
print(f'  After cleaning:               {len(gpvs_clean):,}')
print(f'  After SMOTE:                  {len(X_gpvs_balanced):,}')
print(f'  Features:                     {GPVS_SENSOR_COLS}')
print(f'  Target:                       fault_type (F0-F7, {num_classes} classes)')
print(f'  Balancing:                    SMOTE')
print(f'  Scaling:                      StandardScaler')
print(f'  Split:                        70/15/15 (stratified)')
print(f'  Train: {len(X_train_g):,} | Val: {len(X_val_g):,} | Test: {len(X_t

print('\n[MODEL RESULTS]')
print(f'  XGBoost F1-Score:             {f1:.4f} (target > 0.85)')
print(f'  Linear Learner RMSE:          {model_rmse:.4f} W/m2')
print(f'  Baseline RMSE:                {baseline_rmse:.4f} W/m2')
print(f'  RMSE Reduction:               {rmse_pct:.1f}% (target >= 20%)')

print(f'\nPrepared data: s3://{BUCKET}/prepared/')
print(f'Models:                        s3://{BUCKET}/models/')
print(f'GitHub:                        https://github.com/darth-franky/Applied-Data-Science/
print('=' * 62)
```



```
=====
SOLARIS ANALYTICS – DATA PREP & TRAINING SUMMARY
=====
```

[DATA PREPARATION – NSRDB]

```
Raw rows:          350,640
Cleaned (daytime): 175,131
Features:          ['DNI', 'DHI', 'Temperature', 'Wind_Speed', 'Month_
sin', 'Month_cos', 'Hour_sin', 'Hour_cos', 'DayOfYear']
Target:           GHI (continuous)
Scaling:          StandardScaler
Split:            70/15/15
Train: 122,591 | Val: 26,270 | Test: 26,270
```

[DATA PREPARATION – GPVS]

```
Raw rows:          2,514,120
After cleaning:    2,163,480
After SMOTE:       533,696
Features:          ['Ipv', 'Vpv', 'Vdc', 'ia', 'ib', 'ic', 'va', 'vb',
'vc', 'Iabc', 'If', 'Vabc', 'Vf']
Target:           fault_type (F0–F7, 8 classes)
Balancing:        SMOTE
Scaling:          StandardScaler
Split:            70/15/15 (stratified)
Train: 373,587 | Val: 80,054 | Test: 80,055
```

[MODEL RESULTS]

```
XGBoost F1-Score:  0.9598 (target > 0.85)
Linear Learner RMSE: 76.8173 W/m2
Baseline RMSE:     290.4492 W/m2
RMSE Reduction:    73.6% (target >= 20%)
```

```
Prepared data: s3://solaris-analytics-bucket/prepared/
Models:        s3://solaris-analytics-bucket/models/
GitHub:        https://github.com/darth-franky/Applied-Data-Science/tree/main/ADS508
```

```
=====
In [ ]:
```